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Department of Data Science Engineering

Project Title

HealthWise AI

Hospital Management System with Report Analysis

SOFTWARE REQUIREMENTS SPECIFICATION

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Under the guidance of
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CERTIFICATE

This is to certify that the Software Requirements Specification entitled "HealthWise AI: Hospital Management System with Report Analysis" has been prepared by the following students of the Department of Data Science Engineering, D. Y. Patil School of Engineering and Management, Kolhapur, in partial fulfilment of the requirements for the degree of Bachelor of Technology in Data Science Engineering during the academic year 2026 - 2027.

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To the best of our knowledge, the work is original and has not been submitted elsewhere for any other degree or diploma.

Prof. Ajit Patil
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Date: 12 / 10 / 2026

Place: Kolhapur

DECLARATION

We hereby declare that this Software Requirements Specification is the result of our own work, carried out under the guidance of Prof. Ajit Patil. The requirements, diagrams and tables in this document were prepared by us for the HealthWise AI project. Where we have used ideas, standards or text from other sources, they are listed in the References (section 1.7).

We also declare that the project is meant for education and demonstration. It is not a medical device and does not diagnose disease or prescribe medicine.

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TABLE OF CONTENTS

1. Introduction 1
1.1 Project Overview 1
1.2 Purpose of the System 1
1.3 Scope of the Project 1
1.4 Objectives 2
1.5 Intended Users 2
1.6 Definitions, Acronyms and Abbreviations 3
1.7 References 3
2. Overall Description 5
2.1 Product Perspective 5
2.2 Product Functions 5
2.3 User Classes 5
2.4 Operating Environment 6
2.5 Design and Implementation Constraints 7
2.6 Assumptions and Dependencies 7
3. Functional Requirements 8
3.1 Authentication 8
3.2 User Management 8
3.3 Report Upload and Analysis (Main Module) 9
3.4 Insights for Any Type of Report 9
3.5 Doctor Review and Clinical Notes 10
3.6 Prescriptions and Pharmacy 10
3.7 Appointments 10
3.8 History, Comparison and Trends 11
3.9 Dashboards, Notifications and Audit 11
3.10 Key Numbers and Doctor Attention List 11
4. Non-Functional Requirements 12
4.1 Performance 12
4.2 Security 12
4.3 Usability 12
4.4 Reliability 12
4.5 Availability 13
4.6 Maintainability 13
4.7 Scalability 13
4.8 Portability 13
5. External Interface Requirements 14

5.1 User Interface 14
5.2 Hardware Interface 15
5.3 Software Interface 15
5.4 Communication Interface 15
6. Use Case Requirements 16
6.1 Actors 16
6.2 Use Case Diagram 16
6.3 Use Case Descriptions 18
7. System Models 24
7.1 Context Diagram 24
7.2 DFD 25
7.3 Activity Diagram 27
7.4 Sequence Diagram 28
7.5 Class Diagram 29
7.6 ER Diagram 31
8. Data Requirements 33
8.1 Database Tables 33
8.2 Data Dictionary 33
9. System Constraints 41
10. Acceptance Criteria 42
11. Requirements Traceability Matrix 44
12. Appendices 51
Appendix A: Glossary 51
Appendix B: Sample Forms and Screens 51
Appendix C: Sample Reports 62
Appendix D: Role-Permission Matrix 62
Appendix E: Database Schema 63
Appendix F: Sample Questionnaire 65
Appendix G: References 65

1. Introduction

1.1 Project Overview

HealthWise AI is a web-based hospital management system. A hospital's staff upload a patient's medical report (a laboratory report, an X-ray report, an ECG, a discharge summary and so on) as a PDF, a photo or a text file. The system reads the file, finds the results and explains them in plain language. It then gives the treating doctor a short list of key findings, red flags and suggestions, and it prepares general tips for the patient.

Nothing reaches the patient until the doctor has reviewed it. After the review the doctor can add notes, follow-up advice and a prescription; the pharmacy dispenses the medicine and keeps the stock; the patient can follow the whole story in a timeline and compare reports over time. Five roles use the system: Admin, Doctor, Employee, Patient and Pharmacy Staff, and each role sees only what it needs.

Analysis is rule-based. The system calls a result Within, Above or Below only by comparing it with the normal range printed in the same report. An optional language model may reword the text, but it never decides anything and never sees personal details.

1.2 Purpose of the System

The system is being developed to solve the following problems of a typical hospital:

- Medical reports arrive in many layouts and formats; reading them by hand is slow and error-prone.
- Patients receive numbers they cannot understand, and doctors have little time to explain.
- Reports, notes, prescriptions and dispensing records are kept in separate places, so the patient's story is hard to follow.
- Patient information must be seen only by the right people, and there must be a record of who did what.

The purpose of this SRS is to state clearly what the system shall do, so that the development team, the guide and the testers share one understanding, and so that every requirement can be traced to a use case and a test.

1.3 Scope of the Project

Included:

- Login for five roles, forced password change and role-based access.
- Staff creation, department management and patient registration.
- Report upload (PDF, JPG, JPEG, PNG, WEBP, TXT, CSV), text reading and OCR.
- Rule-based analysis of laboratory results using the printed reference range.

- Insights for any type of medical report: key findings, red flags, suggestions for doctors and tips for patients.
- Mandatory doctor review before the patient sees an analysis; notes, advice and medical history.
- Prescriptions, pharmacy dispensing and medicine stock.
- Appointments, patient timeline, report comparison and trend charts.
- Dashboards, in-app notifications and an audit log.

Not included:

- Diagnosis or prediction of disease, and any choice of medicine by the system.
- Billing, insurance and payroll.
- Tele-consultation (video calls), ward or bed management.
- Email and SMS messages (notifications are shown inside the application).
- Exchange of data with other hospital systems (for example HL7 or FHIR).
- Native mobile apps (the web application works in a phone browser).

1.4 Objectives

1. To read uploaded reports of different layouts and formats and to extract real values from each file.
2. To classify each result using only the range printed in the report, so that no range is ever invented.
3. To give doctors insights and suggestions for any type of report, and give patients clear tips, within seconds for a text report.
4. To make sure that a doctor reviews every report before the patient sees its analysis (100% of reports).
5. To connect report, review, notes, prescription and dispensing in one workflow.
6. To provide role-based access so that each user sees only permitted data, and to record important actions in an audit log.
7. To show the patient's history over time with comparison and trend charts.
8. To keep the system verifiable: all 108 automated tests pass.

1.5 Intended Users

- Administrator: manages staff accounts and reads the audit log.
- Doctor: reviews reports, writes notes and prescriptions.
- Hospital Employee: registers patients, uploads reports, books appointments.
- Patient: views reviewed reports, tips, prescriptions and history.
- Pharmacy Staff: dispenses medicine and manages stock.
- Readers of this document: the project guide, faculty reviewers, developers and testers.

1.6 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
Admin	System Administrator
UI	User Interface
DB	Database
API	Application Programming Interface
REST	Representational State Transfer, a style of web API that exchanges JSON over HTTP
JWT	JSON Web Token, the signed token a user receives after logging in
RBAC	Role-Based Access Control: what a user may see or do depends on the user's role
OCR	Optical Character Recognition: reading text from a scanned page or a photo
LLM	Large Language Model: an optional AI service used only to reword text
Reference range	The normal range printed next to a result in the original report
Insight	A plain-language statement the system produces about a report: key findings, red flags, doctor suggestions and patient tips
Red flag	A statement in a report that uses urgent wording and is therefore shown to the doctor first
FR / NFR	Functional Requirement / Non-Functional Requirement
UC	Use Case
DFD	Data Flow Diagram
ER	Entity-Relationship
RTM	Requirements Traceability Matrix
HTTP / HTTPS	Hypertext Transfer Protocol / its encrypted form
PDF	Portable Document Format
SPA	Single-Page Application

1.7 References

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12. Project documents: docs/ARCHITECTURE.md, docs/PROJECT_REPORT.md and README.md in the HealthWise AI repository.

2. Overall Description

This chapter gives a high-level view of the complete system.

2.1 Product Perspective

HealthWise AI is a new, self-contained system. It replaces the manual way of keeping paper reports, reading numbers by eye and passing notes and prescriptions between desks. It is not an extension of an older software product.

The system has three tiers: a React web application in the user's browser, a FastAPI server that holds all business rules, and a relational database (SQLite or PostgreSQL). Two outside services are used and both are replaceable: the Tesseract OCR engine (to read scans) and an optional language model (to reword text). Figure 7.1 shows the system and its surroundings.

2.2 Product Functions

The major functions of the system are summarised below. Chapter 3 describes each in detail.

Function group	Summary
Accounts and access	Login, forced password change, five roles, role-based access, staff creation, activate / deactivate.
Patients and appointments	Register and search patients, book appointments with clash checks, confirm / complete / cancel.
Report upload and reading	Upload PDF, photo or text; read text or run OCR; find tests, values, units and printed ranges.
Rule-based analysis	Mark every result Within, Above, Below or Unknown using only the range printed in the report.
Any-report insights	For lab, imaging, ECG, pathology, discharge, prescription and clinical-note reports: key numbers at a glance, key findings, red flags, suggestions for doctors and tips for patients.
Doctor review	Approve, modify or ask for clarification; notes, recommendations and medical history. Patients see results only after this step.
Prescriptions and pharmacy	Doctor writes prescriptions; pharmacy dispenses in part or in full, marks medicines unavailable and keeps stock.
History and trends	Searchable patient timeline, side-by-side report comparison and trend charts.
Dashboards and audit	A dashboard for each role built from live data (for the doctor, a list of waiting reports that need attention first) and an audit log of important actions for the administrator.

2.3 User Classes

User	Responsibilities
Admin	Creates and deactivates staff accounts, manages departments, reads the audit log. Never sees clinical values.

User	Responsibilities
Doctor	Reviews reports and the system's suggestions, writes notes, advice and prescriptions, sees authorised patients' history.
Hospital Employee	Registers patients, uploads reports, books appointments. Never sees clinical values.
Patient	Views own reviewed reports, tips, notes, prescriptions, appointments, timeline and trends.
Pharmacy Staff	Opens active prescriptions, dispenses medicine, records unavailable items, manages stock.

Appendix D gives the full role-permission matrix.

2.4 Operating Environment

Hardware

Item	Server (minimum)	Client (minimum)
Processor	2 cores	Dual-core, any modern CPU
RAM	4 GB (8 GB if OCR is used often)	4 GB
Storage	10 GB plus space for uploaded reports	Not needed
Internet connection	Needed only for the optional AI model and for access from other computers	Network access to the server

Software

Component	Technology
Frontend	React 18, Vite 6, Tailwind CSS 4, React Router 6, Axios, Recharts
Backend	Python 3, FastAPI, Pydantic v2, SQLAlchemy 2
Security	JWT (PyJWT) and bcrypt password hashing
Document reading	pypdf and pypdfium2 for PDFs; Pillow and Tesseract (pytesseract) for OCR
Database	SQLite for development; PostgreSQL for production
Optional AI	Any HTTP-based language model, used only to reword text
Testing	pytest (108 automated tests)
Browser	Google Chrome, Microsoft Edge, Firefox (current versions)
Operating system	Windows 10/11 or Linux (server and client); any OS with a modern browser (client)

2.5 Design and Implementation Constraints

- The system must use a relational database (SQLite for development, PostgreSQL for production).
- Only authenticated users can use the application; every private function needs a valid token.
- Passwords must not be stored as plain text.
- The application must work on current versions of Chrome, Edge and Firefox.
- Reference ranges must come only from the report; the system must not invent medical values.
- The system must not diagnose, predict disease or choose medicine; AI text must pass a safety filter.
- Scanned files can be read only if the Tesseract OCR engine is installed on the server.

2.6 Assumptions and Dependencies

- Reports contain results with numbers and units, and usually print a reference range.
- Doctors check the system's output before the patient sees it; the system supports the doctor and does not replace the doctor.
- The hospital creates all accounts; there is no public registration.
- Users have network access to the server, and the database server is available.
- Patient information entered at registration is correct.
- The optional AI model and Tesseract are separate services; the system works without the AI model.

3. Functional Requirements

The requirements are divided by module. Each requirement has a unique ID, which is used again in the use cases, acceptance criteria and the traceability matrix (chapters 6, 10 and 11).

3.1 Authentication

FR-01: The system shall allow registered users to log in with email and password.

FR-02: The system shall validate the entered credentials and show a general error message, which does not reveal which part was wrong, for invalid credentials.

FR-03: The system shall issue a signed token at login and shall refuse every private request that has no valid token.

FR-04: The system shall allow users to log out, which ends the session in the browser.

FR-05: The system shall have no public sign-up. The first administrator can be created only once, on an empty system.

FR-06: The system shall refuse login, and stop honouring old tokens, for deactivated users.

FR-07: The system shall store passwords only as bcrypt hashes and shall force a password change at first login when the password was generated by the system.

FR-08: The system shall let a logged-in user change their own password after entering the current one.

3.2 User Management

FR-09: The administrator shall be able to create doctors, employees and pharmacy staff, and the system shall show the generated password once.

FR-10: The administrator shall be able to manage departments (create and list).

FR-11: The administrator shall be able to activate or deactivate a user, but not their own account.

FR-12: Employees and the administrator shall be able to register a patient (the login account is created together with the patient) and edit only basic information.

FR-13: The system shall allow searching patients by name, patient code, phone number or email.

FR-14: The system shall limit every function to its roles and shall answer others with the message 'You do not have permission to access this information.'

FR-15: The system shall keep patient data private: a patient sees only their own data, a doctor only authorised patients, and employees and the administrator never see clinical values.

FR-16: The administrator shall be able to view a readable list of all people in the system.

3.3 Report Upload and Analysis (Main Module)

FR-17: The system shall accept PDF, JPG, JPEG, PNG and WEBP files (and optionally TXT and CSV) up to 10 MB, and shall check the real content type, not only the file name.

FR-18: The system shall read the text of text-based PDFs directly, without OCR.

FR-19: The system shall use OCR for scanned PDFs and images, and shall show a clear message when OCR is not available.

FR-20: The system shall extract test name, value, unit and reference range from the real file, whatever the layout, and shall keep the original test name.

FR-21: The system shall take reference ranges only from the report (forms a-b, below x, above x) and shall never invent a range.

FR-22: The system shall decide Within, Above, Below or Unknown by fixed rules, and Unknown when no range is printed.

FR-23: The system shall reject unreadable files and non-medical documents with a clear message, and shall save nothing in that case.

FR-24: The system shall save an accepted report as 'Waiting for doctor' and show it on the assigned doctor's dashboard.

FR-25: The system shall warn when the name printed on the report does not match the selected patient.

FR-26: A doctor shall be able to upload a report for an authorised patient; when the patient has no assigned doctor the employee must choose one.

FR-27: The system shall write explanations that do not diagnose or prescribe, pass them through a safety filter, and fall back to built-in text when no AI model is configured.

FR-28: The system shall keep original files private and give them only to authorised users.

3.4 Insights for Any Type of Report

FR-29: The system shall accept other medical documents (imaging, ECG, pathology, discharge summary, prescription, clinical note), detect their type, and reject non-medical documents such as invoices.

FR-30: For every report the system shall produce key findings (normal or worth a look, with negation such as 'no evidence of' handled), red flags for urgent wording, suggestions for the doctor and general tips for the patient.

FR-31: For laboratory reports the insights shall be grouped by body system and use only printed ranges; a value without a printed range shall not be judged.

FR-32: Doctor-only content (suggestions, red flags) shall be shown only to the doctor; patients shall see general tips only after the doctor's review; employees and administrators shall see none.

FR-33: A doctor shall be able to regenerate the insights of a report (recorded in the audit log); older reports without saved insights shall get them when opened.

FR-34: An optional AI model may improve the wording only after personal details are removed; its answer shall pass the safety filter, otherwise the rule-based text is used.

3.5 Doctor Review and Clinical Notes

FR-35: The doctor shall be able to review a report (approve, modify or ask for clarification); a comment is required for modify and clarify; the report is then marked Reviewed.

FR-36: The system shall show a report's analysis to the patient only after the doctor has reviewed it.

FR-37: The doctor shall be able to add notes (Normal, Important, Follow-up required) and recommendations (with an optional follow-up date); patients shall not be able to edit them.

FR-38: The doctor shall be able to add medical-history entries and shall see the patient's full history before reviewing.

3.6 Prescriptions and Pharmacy

FR-39: Only doctors shall create prescriptions (medicine, dosage, frequency, duration, quantity); a doctor shall see only their own prescriptions.

FR-40: Pharmacy staff shall see prescriptions (cancelled ones are hidden) with only the data they need, and shall never create or edit a prescription.

FR-41: Pharmacy staff shall be able to dispense in part or in full; the system shall not allow more than the remaining quantity, shall block cancelled or expired prescriptions, and shall lower stock for tracked medicines (earliest expiry first, never expired stock) and refuse when the stock is not enough.

FR-42: Pharmacy staff shall be able to mark a medicine as unavailable, and the system shall record it.

FR-43: Pharmacy staff shall be able to manage medicine stock, and the dashboard shall show low-stock and soon-to-expire items.

FR-44: The patient shall be able to see the status of each prescription and how much has been dispensed.

3.7 Appointments

FR-45: The system shall let the administrator, employees and patients book appointments, list them per role, and update their status (the doctor, employee and administrator may confirm, complete or cancel; a patient may only cancel); it shall refuse past times and clashes.

3.8 History, Comparison and Trends

FR-46: The system shall build a searchable patient timeline from the database (appointments, reports, analysis, reviews, notes, recommendations, prescriptions, dispensing, history) and show the source of each item.

FR-47: The system shall compare two reports of one patient (previous, latest, change) with a trend that understands the direction of each parameter.

FR-48: The system shall draw trend charts of each parameter over time.

FR-49: A patient's comparison shall use only reviewed reports.

3.9 Dashboards, Notifications and Audit

FR-50: Each role shall have a dashboard computed from the database, with honest empty states.

FR-51: The system shall show in-app notifications on the dashboards: reports waiting for review, upcoming appointments, low stock and expiring medicine. (Email and SMS are not part of this version.)

FR-52: The system shall write important actions (login, creation, upload, analysis, review, notes, prescriptions, opening a report file or a prescription, dispensing, user changes) to an audit log that only the administrator can read, in plain sentences.

FR-53: The system shall display a medical disclaimer on every page and in the health response of the API.

FR-54: The system shall upgrade an existing database automatically at start-up when new columns are added, keeping the data.

3.10 Key Numbers and Doctor Attention List

FR-55: For a laboratory report the system shall show, at the top of the report, the number of tests read, in range, outside the range and without a range, and a card for each value outside the printed range with its difference from the limit, largest first.

FR-56: For other reports (Holter, ECG, scans and so on) the system shall read the label and number pairs of the text (counts, measurements, values with a printed range) and show them as tiles; it shall mark a tile only when the report prints a range for it or when it counts events above zero, shall leave out personal fields such as age, date or phone, and shall show each number once.

FR-57: The key findings of a report shall be real sentences only: an abbreviation such as 'No.' shall not end a sentence, table rows shall not appear as sentences, and a repeated sentence shall appear once.

FR-58: The doctor's dashboard shall show how many waiting reports have urgent wording and how many values are outside the range, and a 'look at these first' list of the waiting reports with the most important values pulled out, ordered by urgent wording and then by values outside the range.

4. Non-Functional Requirements

These describe how well the system should work, rather than what functions it performs.

4.1 Performance

Target values for a normal server (2 CPU cores, 4 GB RAM):

- The system should answer normal page and list requests within 3 seconds.
- A text-based report (up to 10 MB) should be read, analysed and saved within 10 seconds.
- OCR of a scanned page may take longer; the screen shows that work is in progress and never freezes.
- Lists of people, reports and appointments are paged or limited so that screens stay fast as data grows.

4.2 Security

- Passwords shall be stored only as salted bcrypt hashes, never as plain text.
- Every private request shall carry a signed token; a deactivated user's token stops working at once.
- Access shall be role-based, and a second check shall confirm that the record belongs to the user (for example, a patient only sees their own record).
- Uploaded files shall be checked by real content type and size, stored privately and served only to authorised users.
- Personal details shall be removed before any text is sent to the optional AI model.
- Important actions shall be written to an audit log.

4.3 Usability

- The interface shall use plain words for non-technical users (for example 'Waiting for doctor' instead of status codes) and show dates as 'Today' or '2 hours ago'.
- Important tasks (upload a report, book an appointment, dispense medicine) shall need only a few clicks from the dashboard.
- Error messages shall say what went wrong and what to do next.
- Screens shall work on desktop, tablet and phone widths.

4.4 Reliability

- The system shall keep data consistent: a failed upload saves nothing, and a database change is committed or rolled back as a whole.
- The system shall never fall back to demo or invented data when something fails; it shall show a clear error.
- The database can be backed up with the standard tools of SQLite or PostgreSQL.

- Unexpected errors shall be answered with a typed, readable message and not a crash.

4.5 Availability

The system should be available during hospital working hours, with a target of 99% availability in a month for a production deployment. Planned maintenance should be announced in advance. The optional AI model and OCR engine are not required for the system to stay available: the system continues with built-in text and shows a clear message for scanned files when OCR is missing.

4.6 Maintainability

- The code shall be modular: routers (web layer), services (business rules), database models and utilities are separate.
- Supported lab parameters shall be defined in one catalogue file, so a new test can be added without changing the parser.
- A change to one module shall have minimal effect on others; 108 automated tests protect against regressions.
- Documentation shall be kept in the docs folder and in the README.

4.7 Scalability

The system should support a growing number of patients, reports and users without a change of architecture. The same code runs on SQLite for small installations and on PostgreSQL for larger ones, and the web and database tiers can be moved to separate servers. Uploaded files are kept in a configurable folder that can be moved to shared storage.

4.8 Portability

- The web application shall run on current versions of Chrome, Edge and Firefox.
- The server shall run on Windows or Linux and can be packaged with the supplied Dockerfile.
- Settings (database address, secret key, AI model, file limits) shall come from environment variables.

5. External Interface Requirements

5.1 User Interface

The user interface is a responsive web application in a calm teal theme, written for non-technical users. The main screens are listed below. Screenshots of each screen can be inserted in Appendix B.

Screen	What it shows
Login page	Email and password; first-time setup of the administrator on an empty system.
Change password	Required after a generated password is used.
Admin dashboard	People counts, recent activity, quick actions.
Staff and users	Create doctors, employees and pharmacy staff; activate or deactivate; departments.
Audit log	A day-by-day feed of who did what, in plain sentences.
Employee dashboard	Register patient, upload report and appointment shortcuts.
Register patient	Form for the patient's basic details.
Upload report	Choose patient (and doctor), choose file, see the result or a clear error.
Doctor dashboard	Look-at-these-first list with the important values, reports waiting for review, today's appointments, patients.
Report review	Key numbers at a glance, result cards with range gauges, key findings, red flags, suggestions, history, review form.
Patient details (doctor)	History, reports, notes, recommendations, prescriptions, timeline, trends.
Notes, advice and prescriptions	Cards for the doctor's notes, follow-up advice and prescriptions.
Patient dashboard and My reports	Reviewed reports in plain language, tips and status chips.
Health history and trends	Timeline, report comparison and charts.
Appointments	Cards grouped by day, with date tile, countdown chip and status.
Pharmacy dashboard	Open prescriptions, low stock, expiring medicines.
Dispensing and inventory	Dispense in part or in full, mark unavailable, stock meters, dispensing history.

Common rules for all screens: every page shows the medical disclaimer; status words are written in plain language; dates appear as Today, Tomorrow or 2 hours ago; and long lists are shown as cards or grouped rows instead of dense tables.

5.2 Hardware Interface

The system needs no special hardware. A user may take a photo of a printed report with a phone camera or use an ordinary scanner and then upload the file. No fingerprint reader, RFID reader, IoT device or GPS device is used.

5.3 Software Interface

Software	Use
SQLite / PostgreSQL	Stores all data (SQLAlchemy 2 is the access layer).
Tesseract OCR (pytesseract)	Reads text from scanned PDFs and images.
pypdf, pypdfium2, Pillow	Read PDF text, render PDF pages and prepare images.
Optional language model (HTTP API)	Rewords insights after personal details are removed; the answer is safety-filtered.
Browser	Runs the React application.

No payment gateway, email service, SMS service or map service is used in this version.

5.4 Communication Interface

- The browser and the server talk over HTTP (HTTPS in production) using a REST API with JSON, under the path /api. The API documents itself at /docs.
- Each private request carries a JWT in the Authorization header.
- File uploads use multipart form data.
- In development the Vite server forwards /api to the backend, so no cross-origin settings are needed.
- The server reaches the database and the optional AI model over TCP/IP. No email or SMS communication is used.

6. Use Case Requirements

Use cases are the link between a requirement and its test, so they are written step by step.

6.1 Actors

Actor	Who	Role in the system
Admin	Hospital administrator.	Creates staff accounts, manages departments, reads the audit log; has no access to clinical values.
Doctor	Treating doctor.	Reviews reports and insights, writes notes, advice and prescriptions.
Employee	Front-desk or records staff.	Registers patients, uploads reports, books appointments; has no access to clinical values.
Patient	Person receiving care.	Sees own reviewed reports, tips, prescriptions, appointments and history.
Pharmacy Staff	Pharmacist or assistant.	Dispenses prescribed medicine and keeps stock.
OCR engine (system actor)	Tesseract.	Reads text from scans and photos when the system asks.
AI model (system actor, optional)	External language model.	Rewords text after personal details are removed.

6.2 Use Case Diagram

Figure 6.1 shows the use case diagram. Each actor has its own coloured lane; use cases shared by all actors (login, change password, dashboard) are shown in the dashed box.

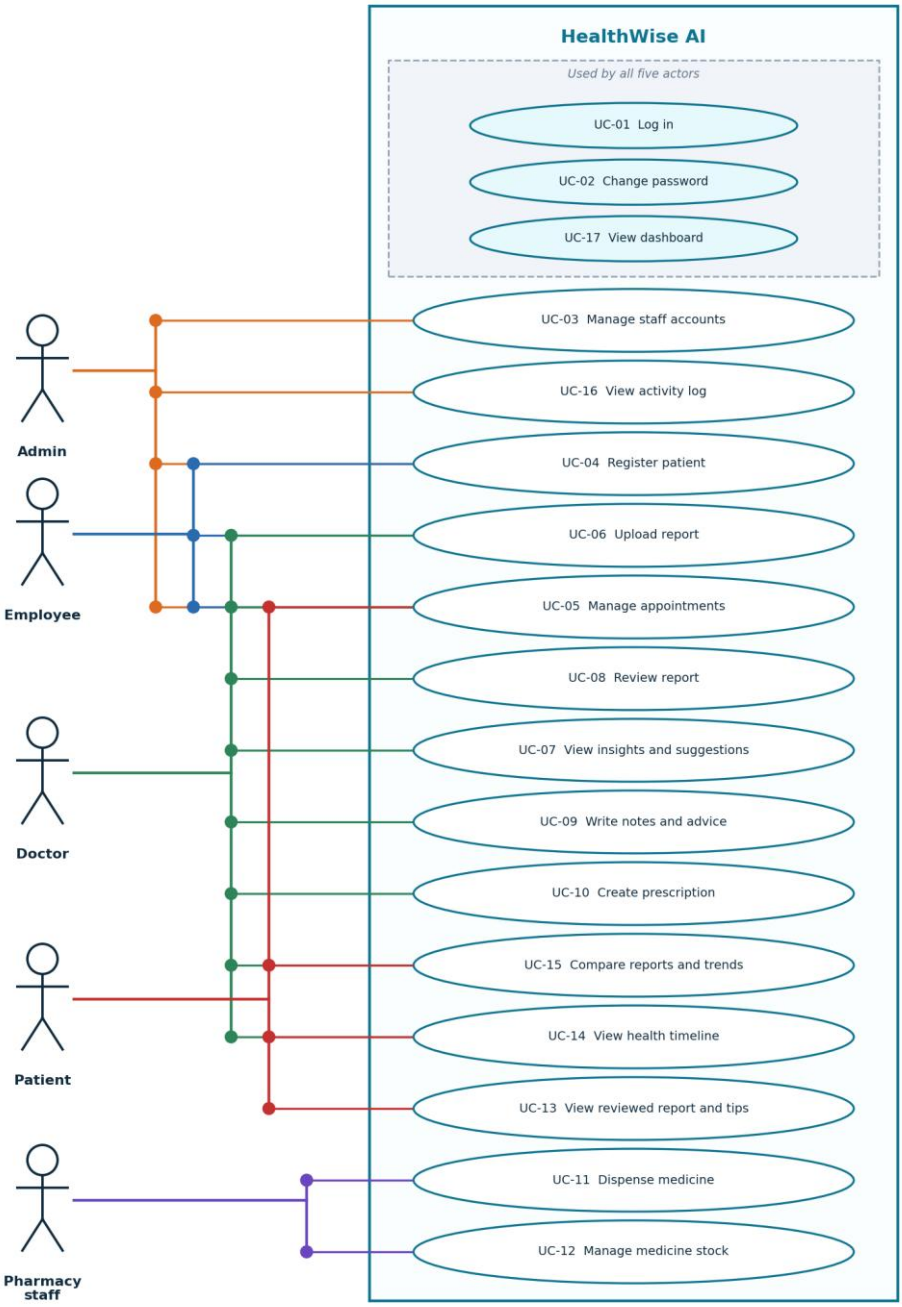


Figure 6.1 Use case diagram

6.3 Use Case Descriptions

UC-01 Login

Use Case ID	UC-01
Use Case	Login
Actor	Admin / Doctor / Employee / Patient / Pharmacy Staff
Precondition	The user has an active account.
Main Flow	Enter email, then password, then the system validates them, then the dashboard of the user's role opens.
Alternative Flow	Wrong credentials or deactivated account: a general error is shown and nothing is revealed. If the password was generated, the user is sent to Change password first.
Postcondition	The user is logged in; the login is written to the audit log.

UC-02 Change password

Use Case ID	UC-02
Use Case	Change password
Actor	All users
Precondition	The user is logged in.
Main Flow	Open Change password, then enter the current and the new password, then the system checks them and saves the new hash.
Alternative Flow	Wrong current password or a new password shorter than 8 characters: an error is shown and nothing changes.
Postcondition	The new password is active and the 'must change' flag is cleared.

UC-03 Manage staff accounts

Use Case ID	UC-03
Use Case	Manage staff accounts
Actor	Admin
Precondition	The administrator is logged in.
Main Flow	Open Staff, then choose a role, then enter name, email and role details, then the system creates the account and shows a generated password once. The administrator can also activate or deactivate people and manage departments.
Alternative Flow	Email already used: an error is shown. The administrator cannot deactivate their own account.
Postcondition	The account exists and can log in; the action is audited.

UC-04 Register patient

Use Case ID	UC-04
Use Case	Register patient
Actor	Admin / Employee
Precondition	The user is logged in.
Main Flow	Open Register patient, then fill in the basic details, then the system creates the patient's account and a patient code.
Alternative Flow	Invalid or duplicate email: an error is shown. Employees cannot enter clinical details.
Postcondition	The patient appears in the patient list and can log in.

UC-05 Manage appointments

Use Case ID	UC-05
Use Case	Manage appointments
Actor	Admin / Employee / Doctor / Patient
Precondition	The user is logged in; the patient and doctor exist.
Main Flow	Choose patient, doctor, date, time and reason, then the system checks the time is not in the past and does not clash, then saves it as Scheduled. The doctor, employee or administrator can confirm, complete or cancel it; a patient can only cancel.
Alternative Flow	Past time or clash: the system explains and does not save.
Postcondition	The appointment shows on the dashboards of the people involved.

UC-06 Upload report

Use Case ID	UC-06
Use Case	Upload report
Actor	Employee / Doctor (OCR engine)
Precondition	The user is logged in; the patient exists.
Main Flow	Open Upload report, then choose the patient (and the doctor if needed), then add the file, then the system checks type, size and content, reads the text or runs OCR, finds the results and insights, and saves the report as Waiting for doctor.
Alternative Flow	Wrong file type, too large, unreadable, or not a medical document: a clear message is shown and nothing is saved. OCR missing for a scan: a clear message. Name on the report differs from the patient: a warning.
Postcondition	The report is saved with its results and insights and shows on the doctor's dashboard.

UC-07 View insights and suggestions

Use Case ID	UC-07
Use Case	View insights and suggestions
Actor	Doctor (AI model)
Precondition	A report has been uploaded for one of the doctor's patients.
Main Flow	Open the report, then read the key numbers at a glance, the key findings, the red flags and the suggestions for the doctor. The doctor can regenerate the insights.
Alternative Flow	A report saved before insights existed: they are created when it is opened.
Postcondition	The doctor has the information needed to review; regeneration is audited.

UC-08 Review report

Use Case ID	UC-08
Use Case	Review report
Actor	Doctor
Precondition	A report is Waiting for doctor.
Main Flow	Open the report, read the values, history and insights, then choose approve, modify or clarify, then add a comment (required for modify and clarify), then save.
Alternative Flow	Comment missing for modify or clarify: the system asks for it.
Postcondition	The report is marked Reviewed; the patient can now see it.

UC-09 Write notes and advice

Use Case ID	UC-09
Use Case	Write notes and advice
Actor	Doctor
Precondition	The doctor is logged in and the patient is authorised.
Main Flow	Open the patient or report, then add a note with its priority, or a recommendation with an optional follow-up date, or a medical-history entry. A suggested tip can be turned into official advice.
Alternative Flow	Empty text: validation message.
Postcondition	The note or advice is saved under the doctor's name and shown to the patient (advice and notes).

UC-10 Create prescription

Use Case ID	UC-10
Use Case	Create prescription
Actor	Doctor
Precondition	The doctor is logged in and the patient is authorised.
Main Flow	Open New prescription, then add each medicine with dosage, frequency, duration and quantity, then save.
Alternative Flow	A missing field: validation message. Only the writing doctor may cancel it later.
Postcondition	The prescription is Active for 30 days and appears for pharmacy and the patient.

UC-11 Dispense medicine

Use Case ID	UC-11
Use Case	Dispense medicine
Actor	Pharmacy Staff
Precondition	An active prescription exists.
Main Flow	Search the prescription, then open it, then enter the quantity (part or all), or mark a medicine unavailable, then the system checks the remaining quantity and the expiry, saves the dispensing and lowers stock.
Alternative Flow	More than remaining, or the prescription is cancelled or expired: the system refuses and says why.
Postcondition	The prescription status is updated (partly or fully dispensed); stock is reduced; the action is audited.

UC-12 Manage medicine stock

Use Case ID	UC-12
Use Case	Manage medicine stock
Actor	Pharmacy Staff
Precondition	The user is logged in.
Main Flow	Open Inventory, then add or update a medicine with batch, quantity and expiry date.
Alternative Flow	Negative quantity or missing name: validation message.
Postcondition	Stock is current; low-stock and expiring items show on the dashboard.

UC-13 View reviewed report and tips

Use Case ID	UC-13
Use Case	View reviewed report and tips
Actor	Patient
Precondition	The doctor has reviewed the report.
Main Flow	Open My reports, then choose a report, then read the plain-language results, the doctor's notes and the general tips.
Alternative Flow	Report not yet reviewed: the patient is told it is waiting for the doctor and sees no analysis.
Postcondition	The patient understands the result. Opening the original file is recorded in the audit log.

UC-14 View health timeline

Use Case ID	UC-14
Use Case	View health timeline
Actor	Doctor / Patient
Precondition	The user is allowed to see the patient.
Main Flow	Open Health history, then scroll or search the timeline of appointments, reports, reviews, notes, prescriptions and dispensing.
Alternative Flow	No data yet: an honest empty message.
Postcondition	The user sees the history with the source of each item.

UC-15 Compare reports and trends

Use Case ID	UC-15
Use Case	Compare reports and trends
Actor	Doctor / Patient
Precondition	The patient has at least two reports (reviewed ones for a patient).
Main Flow	Open Compare, then choose two reports, then the system shows the change of each value and a trend chart.
Alternative Flow	Fewer than two reports: the screen explains that a second report is needed.
Postcondition	Changes and trends are displayed in plain words.

UC-16 View activity log

Use Case ID	UC-16
Use Case	View activity log
Actor	Admin
Precondition	The administrator is logged in.
Main Flow	Open Audit log, then read the day-by-day feed of who did what.
Alternative Flow	No actions yet: an empty message.
Postcondition	The administrator can see an accountable record of activity.

UC-17 View dashboard

Use Case ID	UC-17
Use Case	View dashboard
Actor	All users
Precondition	The user is logged in.
Main Flow	Open the dashboard; the system shows numbers, lists and notifications computed from the database for the user's role.
Alternative Flow	No data yet: an honest empty state is shown instead of sample numbers.
Postcondition	The user sees the current state and quick actions.

7. System Models

The diagrams below show the system from different viewpoints. All of them describe the system as it is built.

7.1 Context Diagram

The context diagram shows the whole system as one process, with the people and services that exchange data with it.

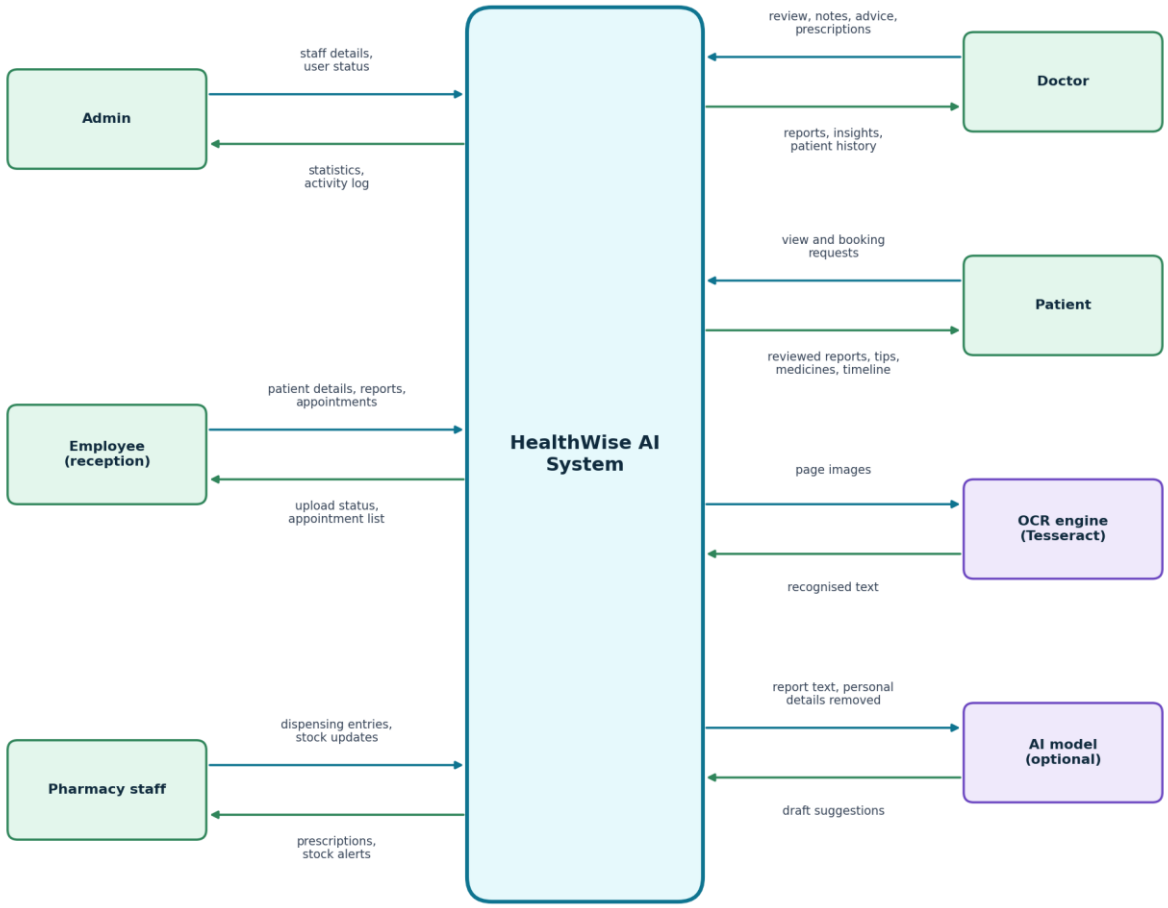


Figure 7.1 Context diagram

7.2 DFD

Level 0 splits the system into six processes and five data stores.

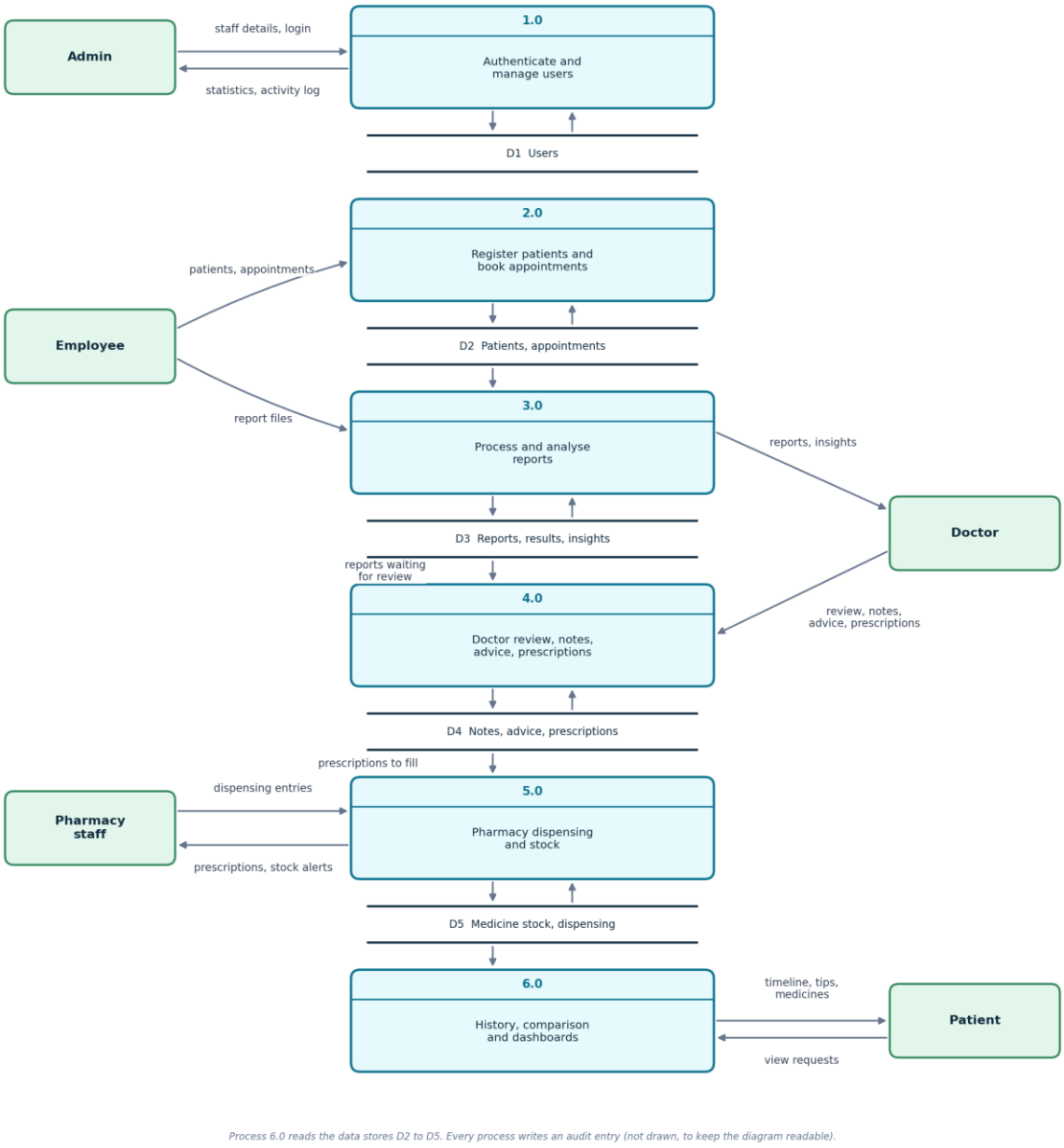


Figure 7.2 Data flow diagram, level 0

Level 1 expands process 3.0 (Report Analysis) into its six steps.

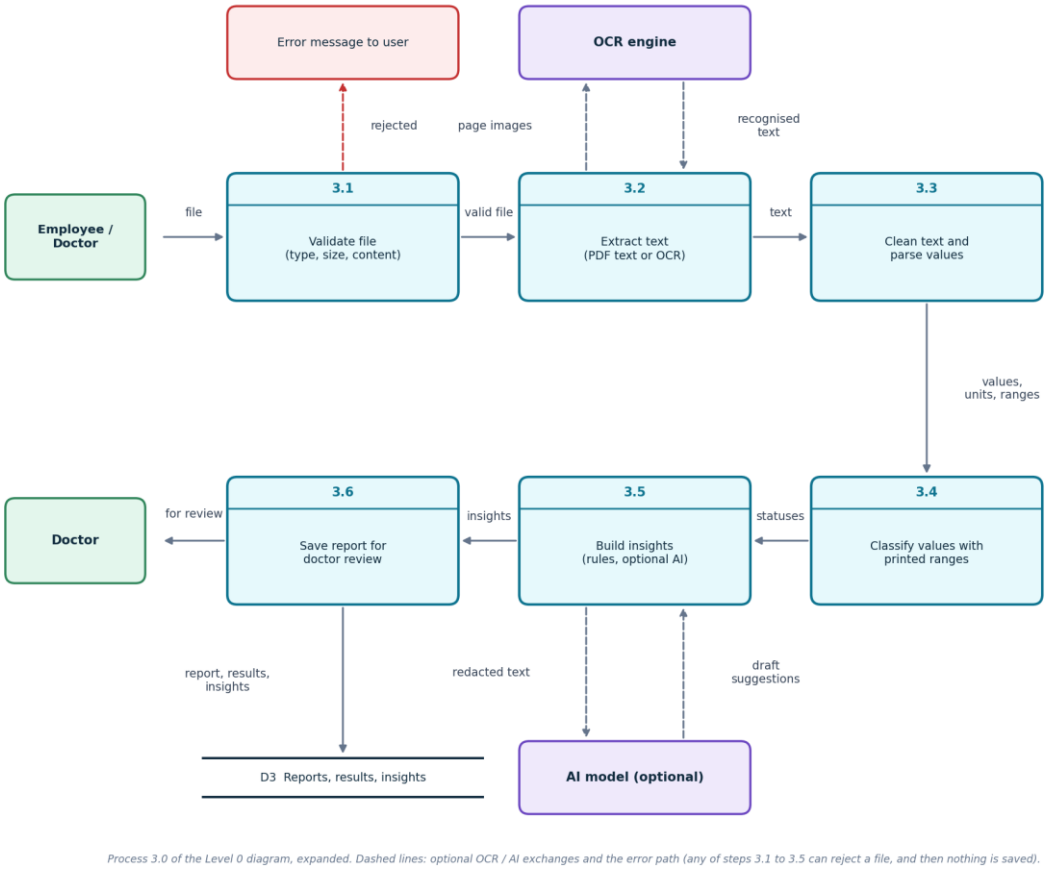


Figure 7.3 Data flow diagram, level 1 (process 3.0, Report Analysis)

7.3 Activity Diagram

The swimlane diagram follows a report from upload by the employee, through the system's checks and the doctor's review, to the patient's view and the pharmacy.

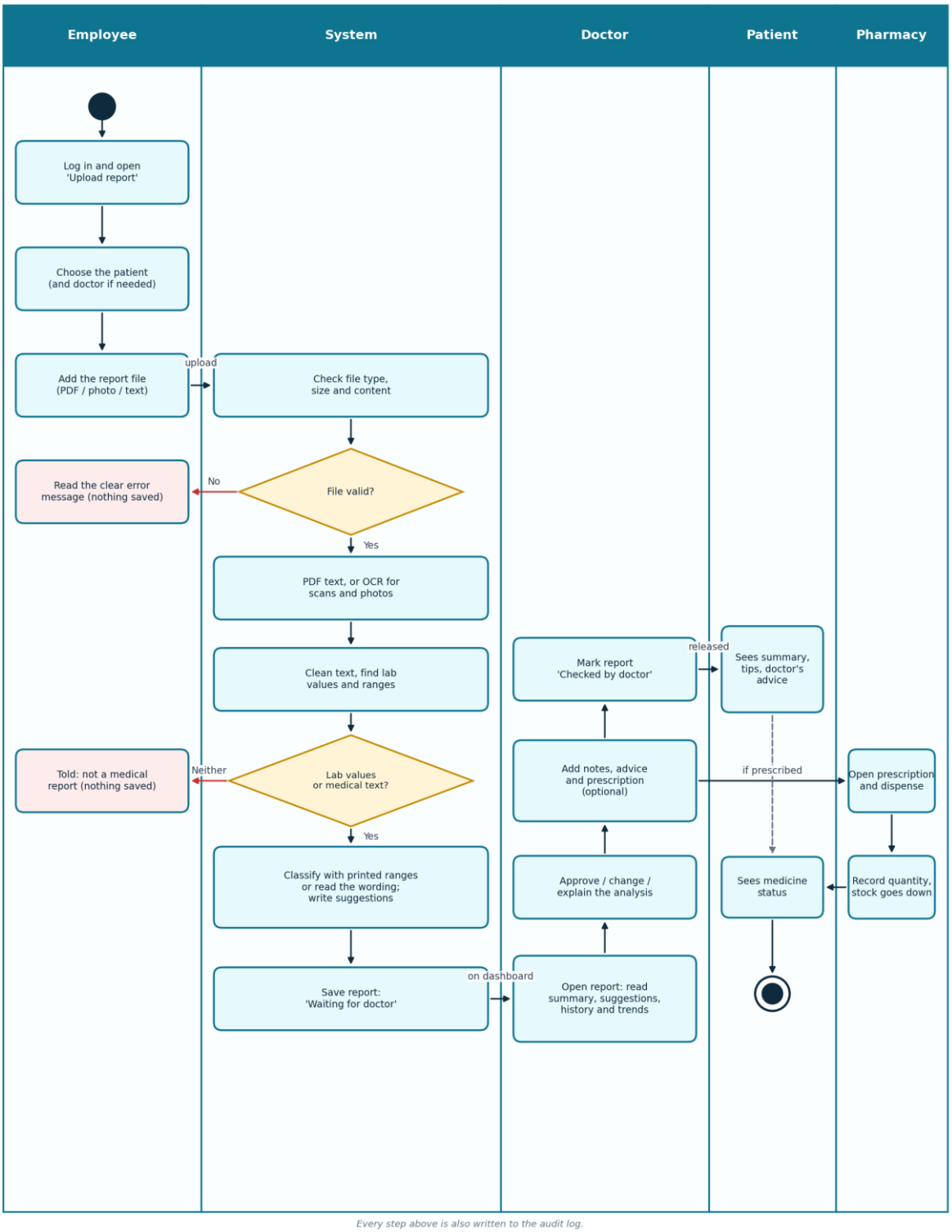


Figure 7.4 Activity diagram: report upload to dispensing

7.4 Sequence Diagram

The sequence diagram shows the messages between the browser, the server, the analysis engine and the database for the same workflow.

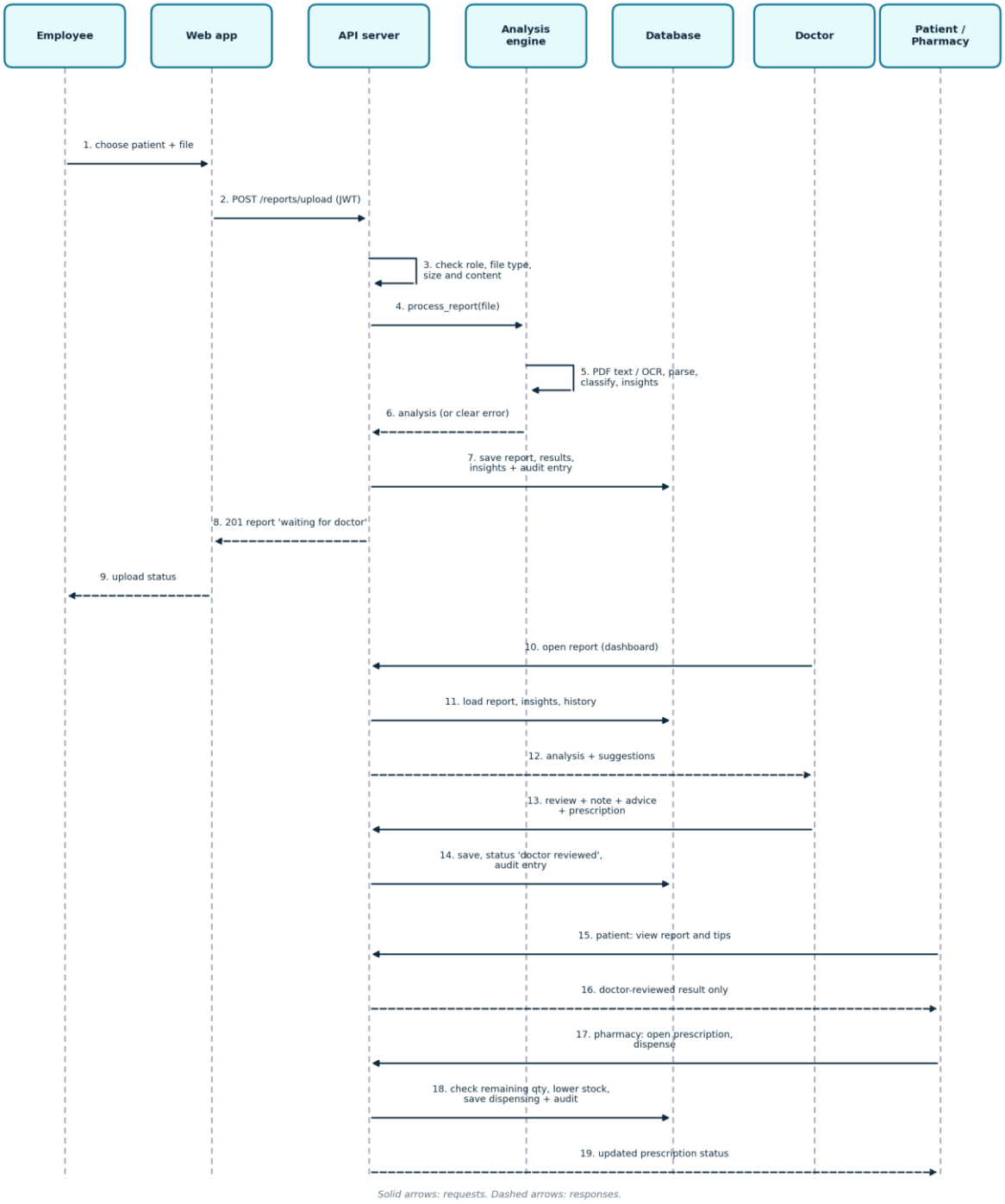


Figure 7.5 Sequence diagram: report upload, review and dispensing

7.5 Class Diagram

The classes are shown in two parts so that the text stays readable on a normal printed page. Part 1 shows the people and their roles; part 2 shows reports, prescriptions and the pharmacy.

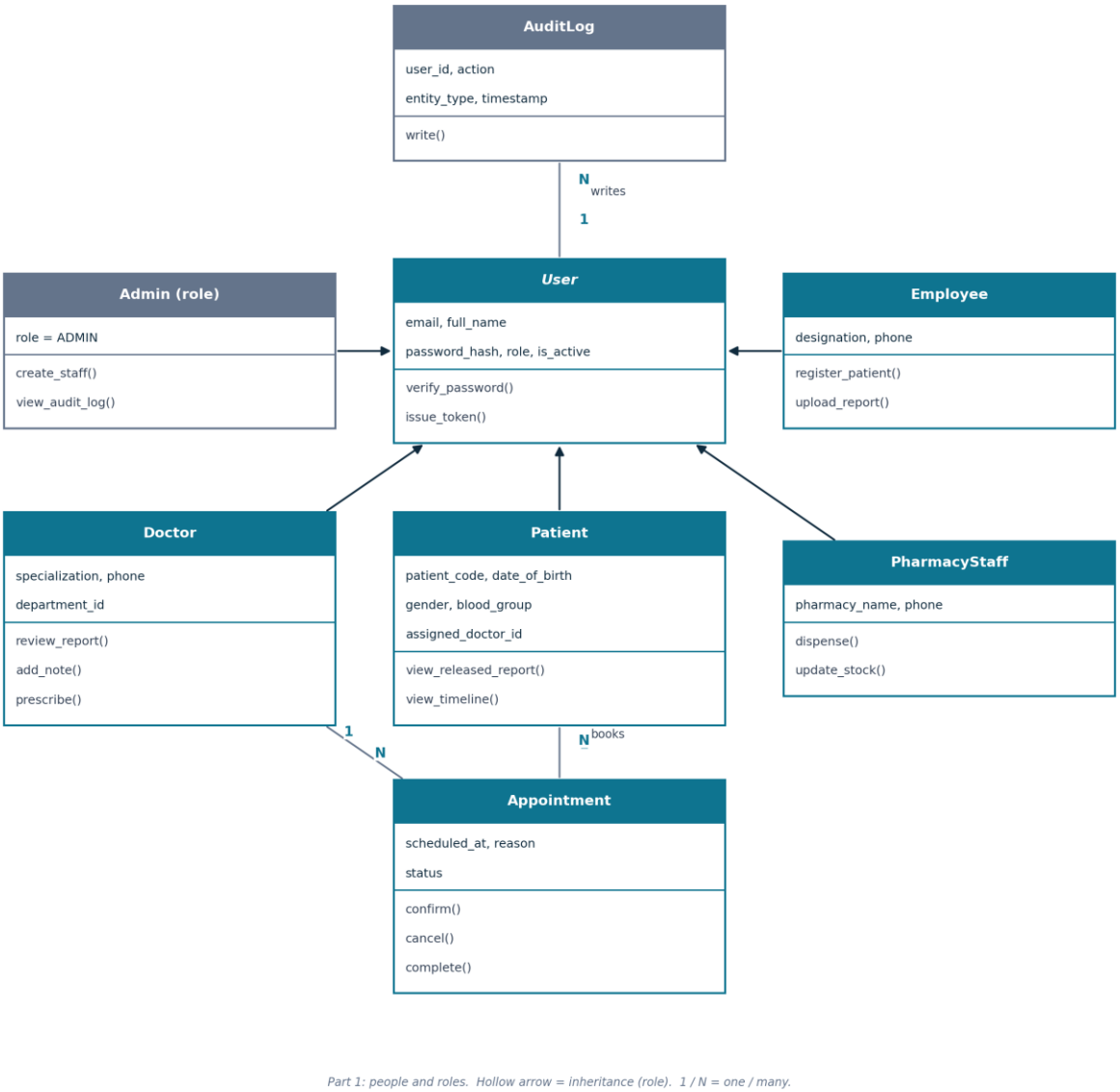
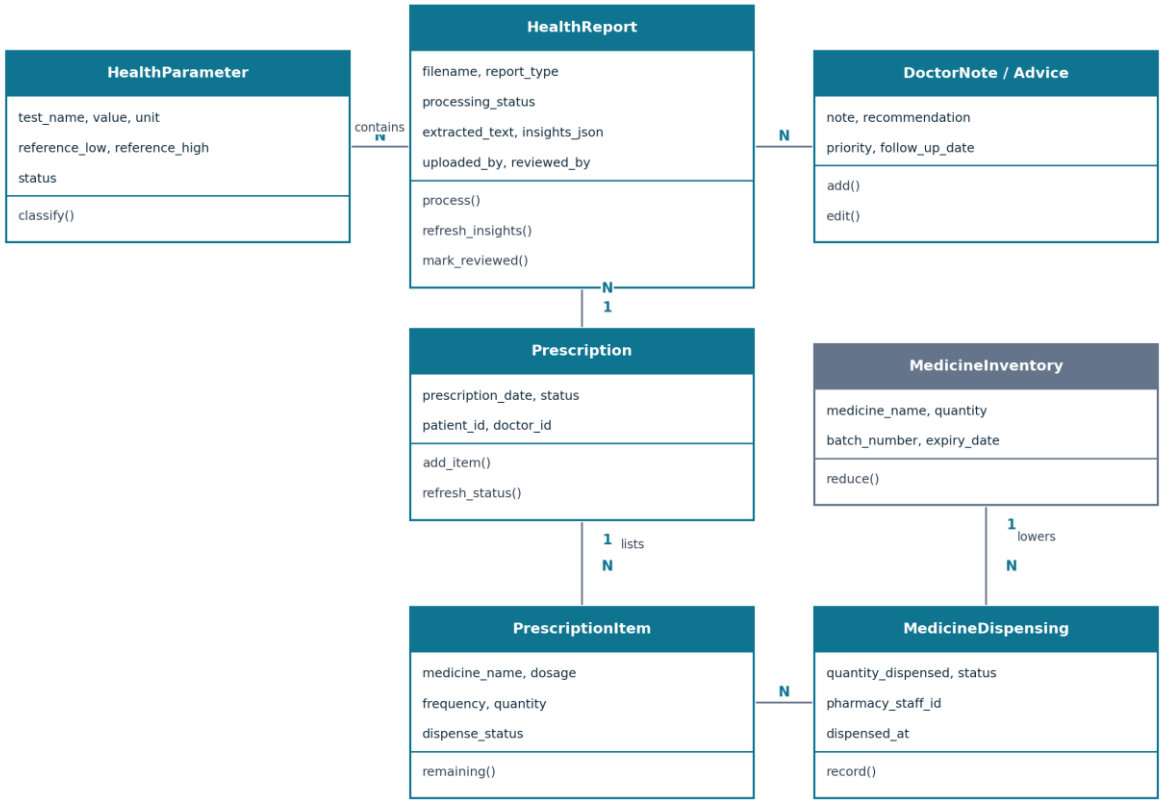


Figure 7.6 Class diagram, part 1: people and roles



Part 2: reports, advice, prescriptions and pharmacy. 1 / N = one / many.

Figure 7.7 Class diagram, part 2: reports, prescriptions and pharmacy

7.6 ER Diagram

The 18 tables are shown in two parts. Only the main relationships are drawn; Chapter 8 lists every column.

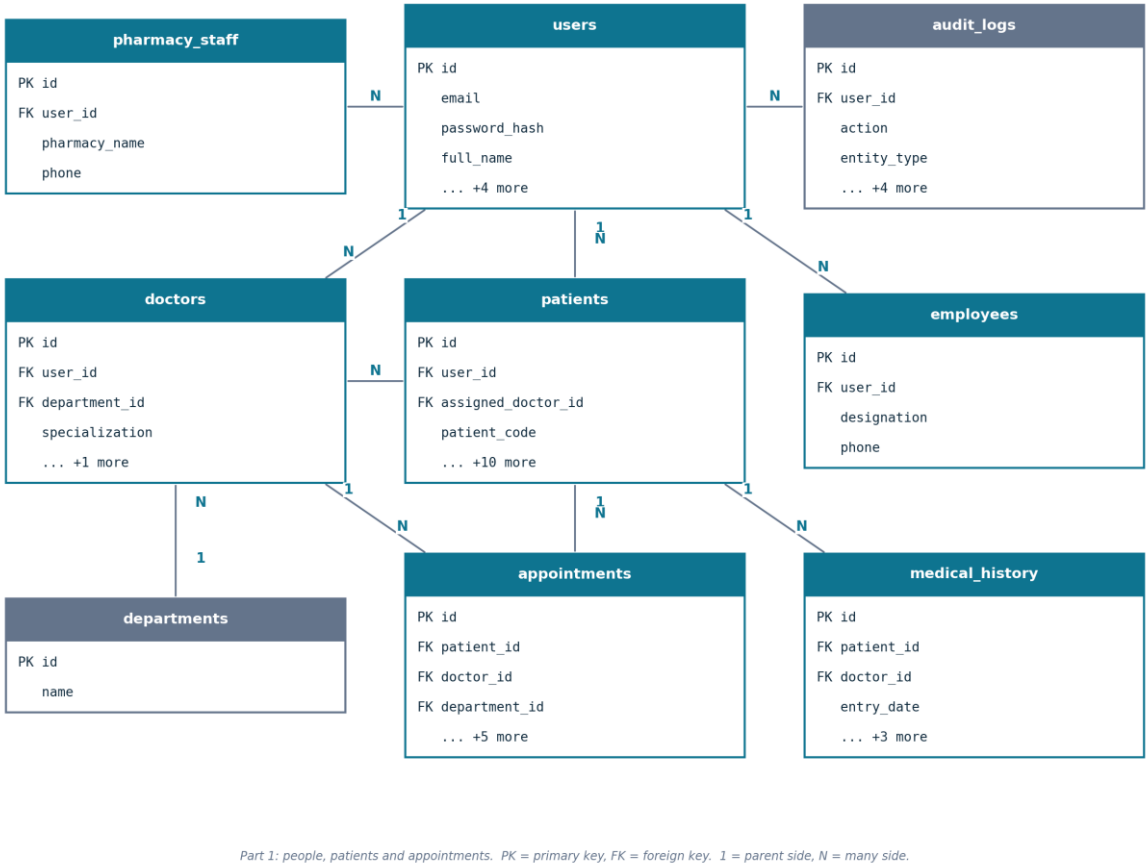


Figure 7.8 ER diagram, part 1: people, patients and appointments

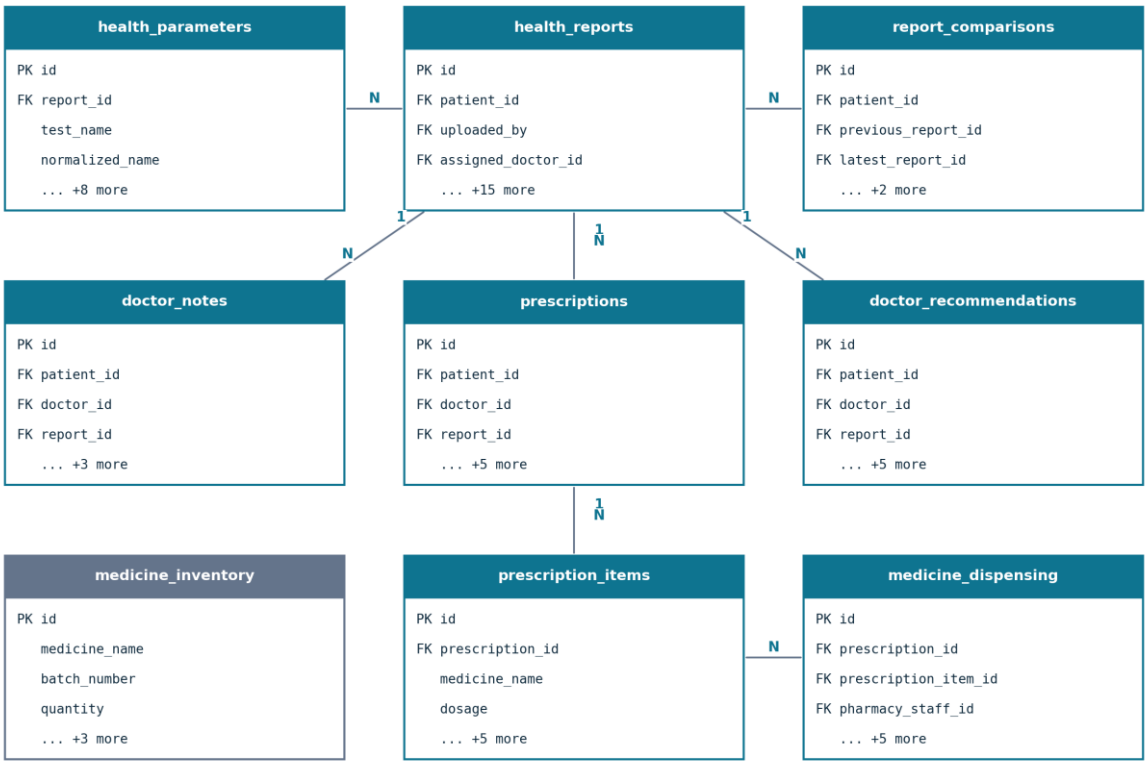


Figure 7.9 ER diagram, part 2: reports, notes, prescriptions and pharmacy

8. Data Requirements

The system stores its data in 18 relational tables. Every table has an integer primary key named id. A foreign key links a row to its parent. The table below gives a short purpose for each table; the data dictionary then lists every column.

8.1 Database Tables

Table	Purpose	Columns
departments	Hospital departments.	2
medicine_inventory	Pharmacy stock of each medicine batch.	7
users	Login accounts of everyone in the system, with the role.	8
audit_logs	Record of important actions.	8
doctors	Extra details of users with the Doctor role.	5
employees	Extra details of users with the Employee role.	4
pharmacy_staff	Extra details of users with the Pharmacy role.	4
patients	Patient profile: demographics and basic history.	14
appointments	Booked visits between a patient and a doctor.	9
health_reports	Uploaded reports, their processing and review state, and the insights.	19
medical_history	Medical-history entries added by a doctor.	7
doctor_notes	Notes written by a doctor.	7
doctor_recommendations	Follow-up advice written by a doctor.	9
health_parameters	One row per test result found in a report.	12
prescriptions	Prescriptions written by a doctor.	9
report_comparisons	Saved comparisons between two reports of a patient.	6
prescription_items	Medicines listed in a prescription.	9
medicine_dispensing	Every handing-out (or 'unavailable' record) of a prescribed medicine.	9

8.2 Data Dictionary

Key: PK = primary key, FK = foreign key (the parent table is named in the description), NULL = may be empty.

8.2.1 users

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
email	VARCHAR(255)	Unique	Login email, unique

Field	Data Type	Key	Description
password_hash	VARCHAR(255)		Password stored as a bcrypt hash
full_name	VARCHAR(150)		Full name
role	VARCHAR(20)		ADMIN, DOCTOR, EMPLOYEE, PATIENT or PHARMACY
is_active	BOOLEAN		False when the account is deactivated
must_change_password	BOOLEAN		True until the generated password is changed
created_at	DATETIME		When the account was created

8.2.2 departments

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
name	VARCHAR(100)	Unique	Department name

8.2.3 doctors

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
user_id	INTEGER	FK	Login account (users)
department_id	INTEGER	FK	Department (departments)
specialization	VARCHAR(120)	NULL	Doctor's speciality
phone	VARCHAR(30)	NULL	Contact number

8.2.4 employees

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
user_id	INTEGER	FK	Login account (users)
designation	VARCHAR(100)	NULL	Job title
phone	VARCHAR(30)	NULL	Contact number

8.2.5 pharmacy_staff

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
user_id	INTEGER	FK	Login account (users)
pharmacy_name	VARCHAR(150)	NULL	Name of the pharmacy

Field	Data Type	Key	Description
phone	VARCHAR(30)	NULL	Contact number

8.2.6 patients

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
user_id	INTEGER	FK	Login account (users)
patient_code	VARCHAR(20)	Unique	Readable patient number, unique
date_of_birth	DATE	NULL	Date of birth
gender	VARCHAR(30)	NULL	Gender
phone	VARCHAR(30)	NULL	Contact number
address	VARCHAR(300)	NULL	Home address
emergency_contact	VARCHAR(150)	NULL	Emergency contact person and number
blood_group	VARCHAR(10)	NULL	Blood group
allergies	TEXT	NULL	Known allergies
medical_history_text	TEXT	NULL	Short history written at registration
current_medications	TEXT	NULL	Medicines currently taken
assigned_doctor_id	INTEGER	FK	Doctor looking after the patient or report (doctors)
created_at	DATETIME		When the patient was registered

8.2.7 appointments

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
patient_id	INTEGER	FK	Patient the record belongs to (patients)
doctor_id	INTEGER	FK	Doctor responsible (doctors)
department_id	INTEGER	FK	Department (departments)
scheduled_at	DATETIME		Date and time of the visit
reason	VARCHAR(300)	NULL	Reason for the visit
status	VARCHAR(20)		SCHEDULED, CONFIRMED, COMPLETED or CANCELLED
created_by	INTEGER	FK	User who created the record (users)
created_at	DATETIME		When the record was created

8.2.8 health_reports

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
patient_id	INTEGER	FK	Patient the record belongs to (patients)
uploaded_by	INTEGER	FK	User who uploaded the file (users)
assigned_doctor_id	INTEGER	FK	Doctor looking after the patient or report (doctors)
filename	VARCHAR(255)		Original file name
file_path	VARCHAR(500)	NULL	Private storage path of the file
report_date	DATE	NULL	Date printed on the report
uploaded_at	DATETIME		When the file was uploaded
processing_status	VARCHAR(20)		ANALYZED (only successfully read reports are stored)
doctor_review_status	VARCHAR(20)		PENDING (waiting for doctor) or REVIEWED
reviewed_by	INTEGER	FK	Doctor who reviewed (doctors)
reviewed_at	DATETIME	NULL	When it was reviewed
review_decision	VARCHAR(20)	NULL	APPROVED, MODIFIED or CLARIFIED
review_comment	TEXT	NULL	Doctor's comment
extracted_text	TEXT	NULL	Text read from the file
processing_method	VARCHAR(20)	NULL	How the text was read: pdf_text, ocr or text
ai_summary	TEXT	NULL	Short plain-language summary
report_type	VARCHAR(30)	NULL	Detected type (lab, imaging, ECG ...)
insights_json	TEXT	NULL	Saved insights (findings, key numbers, red flags, suggestions, tips) as JSON

8.2.9 health_parameters

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
report_id	INTEGER	FK	Report the record relates to (health_reports)
test_name	VARCHAR(200)		Test name as printed
normalized_name	VARCHAR(100)		Standard name of the test
value	FLOAT		Measured value in the report
unit	VARCHAR(50)	NULL	Unit of measure
reference_text	VARCHAR(120)	NULL	Reference range as printed
reference_low	FLOAT	NULL	Lower limit of the range
reference_high	FLOAT	NULL	Upper limit of the range

Field	Data Type	Key	Description
status	VARCHAR(20)		WITHIN_RANGE, ABOVE_RANGE, BELOW_RANGE or UNKNOWN
ai_explanation	TEXT	NULL	Plain-language explanation
created_at	DATETIME		When the record was created

8.2.10 report_comparisons

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
patient_id	INTEGER	FK	Patient the record belongs to (patients)
previous_report_id	INTEGER	FK	Earlier report (health_reports)
latest_report_id	INTEGER	FK	Later report (health_reports)
result_json	TEXT		Comparison result as JSON
created_at	DATETIME		When the record was created

8.2.11 doctor_notes

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
patient_id	INTEGER	FK	Patient the record belongs to (patients)
doctor_id	INTEGER	FK	Doctor responsible (doctors)
report_id	INTEGER	FK	Report the record relates to (health_reports)
note	TEXT		Text of the note
priority	VARCHAR(25)		Normal, Important or Follow-up required
created_at	DATETIME		When the note was written

8.2.12 doctor_recommendations

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
patient_id	INTEGER	FK	Patient the record belongs to (patients)
doctor_id	INTEGER	FK	Doctor responsible (doctors)
report_id	INTEGER	FK	Report the record relates to (health_reports)
title	VARCHAR(200)		Short heading
recommendation	TEXT		Text of the advice
follow_up_date	DATE	NULL	Suggested date of the next visit

Field	Data Type	Key	Description
priority	VARCHAR(25)		Normal, Important or Follow-up required
created_at	DATETIME		When the record was created

8.2.13 medical_history

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
patient_id	INTEGER	FK	Patient the record belongs to (patients)
doctor_id	INTEGER	FK	Doctor responsible (doctors)
entry_date	DATE		Date of the entry
entry_type	VARCHAR(40)		Kind of entry
description	TEXT		Free text
created_at	DATETIME		When the record was created

8.2.14 prescriptions

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
patient_id	INTEGER	FK	Patient the record belongs to (patients)
doctor_id	INTEGER	FK	Doctor responsible (doctors)
report_id	INTEGER	FK	Report the record relates to (health_reports)
prescription_date	DATE		Date written (valid for 30 days)
clinical_indication	VARCHAR(300)	NULL	Why it was prescribed
status	VARCHAR(25)		ACTIVE, PARTIALLY_DISPENSED, DISPENSED, CANCELLED or EXPIRED
notes	TEXT	NULL	Extra notes
created_at	DATETIME		When the record was created

8.2.15 prescription_items

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
prescription_id	INTEGER	FK	Prescription (prescriptions)
medicine_name	VARCHAR(200)		Name of the medicine
dosage	VARCHAR(100)		Dose per intake
frequency	VARCHAR(100)		How often

Field	Data Type	Key	Description
duration	VARCHAR(100)		For how long
quantity	INTEGER		Quantity prescribed
instructions	TEXT	NULL	How to take it
dispense_status	VARCHAR(25)		PENDING, PARTIALLY_DISPENSED, DISPENSED or UNAVAILABLE

8.2.16 medicine_inventory

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
medicine_name	VARCHAR(200)		Name of the medicine
batch_number	VARCHAR(50)	NULL	Batch number
quantity	INTEGER		Units in stock
expiry_date	DATE	NULL	Expiry date
manufacturer	VARCHAR(150)	NULL	Manufacturer
unit_price	FLOAT	NULL	Price per unit

8.2.17 medicine_dispensing

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
prescription_id	INTEGER	FK	Prescription (prescriptions)
prescription_item_id	INTEGER	FK	Medicine line being dispensed (prescription_items)
pharmacy_staff_id	INTEGER	FK	Pharmacy user who dispensed (pharmacy_staff)
quantity_prescribed	INTEGER		Quantity asked for
quantity_dispensed	INTEGER		Quantity handed out
dispensed_at	DATETIME		When it was dispensed
status	VARCHAR(25)		Status of the medicine line after this dispensing
notes	TEXT	NULL	Extra notes

8.2.18 audit_logs

Field	Data Type	Key	Description
id	INTEGER	PK	Unique number of the record
user_id	INTEGER	FK	Login account (users)

Field	Data Type	Key	Description
action	VARCHAR(50)		What was done
entity_type	VARCHAR(50)	NULL	Kind of record touched
entity_id	INTEGER	NULL	Number of the record touched
details	VARCHAR(300)	NULL	Readable details
ip_address	VARCHAR(60)	NULL	Address the request came from
timestamp	DATETIME		When it happened

9. System Constraints

The following limits apply to the project.

- Only registered users can use the system; there is no public sign-up, and the five roles are fixed.
- Uploaded files may be at most 10 MB and must be PDF, JPG, JPEG, PNG, WEBP (optionally TXT or CSV).
- Scanned reports can be read only if the Tesseract OCR engine is installed; otherwise the user gets a clear message.
- Accuracy of reading depends on the quality of the file; a poor scan may be rejected.
- Ranges come only from the report. A value without a printed range is shown as Unknown, never judged.
- The system does not diagnose, predict disease or suggest medicines; it supports the doctor, who stays responsible.
- A prescription is valid for 30 days; a prescription with undispensed medicine after that is treated as Expired.
- The system depends on the availability of the database. SQLite suits a small installation; PostgreSQL is recommended for many users.
- The optional AI model needs an internet connection; without it the system uses its built-in text.
- The project is for education and demonstration and is not certified as a medical device.
- Only authorised doctors can change clinical content; employees and administrators cannot see clinical values.

10. Acceptance Criteria

The system is accepted when all of the following can be shown. The last column links each criterion to the functional requirements of chapter 3.

Requirement	Acceptance Criteria	Related FR
Login	Valid credentials open the dashboard of the user's role.	FR-01
Login	Invalid credentials show a general error and no dashboard.	FR-02
Login	A deactivated user cannot log in and an old token stops working.	FR-06
Security	A user with no valid token is refused (HTTP 401) on every private function.	FR-03
Security	Passwords are stored only as hashes; a generated password must be changed at first login.	FR-07
Roles	A patient sees only their own data; a doctor sees only authorised patients; employees and the administrator never see clinical values.	FR-14, FR-15
Upload	A readable PDF, photo or text report is accepted and saved as 'Waiting for doctor'.	FR-17, FR-24
Upload	A non-medical or unreadable file is rejected with a clear message and nothing is saved.	FR-23, FR-29
Upload	Two different reports give different extracted values.	FR-20
Analysis	Each result is Within, Above or Below only when the report prints a range; otherwise Unknown.	FR-21, FR-22
Insights	Doctors see findings, red flags and suggestions; patients see tips only after review.	FR-30, FR-32
Insights	A normal statement such as 'no evidence of ...' is not flagged as a problem.	FR-30
Review	A report marked Reviewed becomes visible to the patient; before that, the patient sees no analysis.	FR-35, FR-36
Prescription	Only a doctor can create a prescription; pharmacy cannot edit it.	FR-39, FR-40
Dispensing	Dispensing more than the remaining quantity is refused; stock decreases by the dispensed quantity.	FR-41
Dispensing	A cancelled or expired prescription cannot be dispensed.	FR-41
Appointments	A past time or a clashing time is refused.	FR-45
History	The timeline shows the source of every item and can be searched.	FR-46
Comparison	Two reports of a patient show the change of each value and a trend.	FR-47, FR-48
Audit	Important actions appear in the audit log, readable by the administrator only.	FR-52
Quality	All 108 automated tests pass.	All
Key numbers	A lab report shows the values outside the range first; a Holter or ECG report shows its numbers as tiles, with no broken sentences.	FR-55, FR-56, FR-57

Requirement	Acceptance Criteria	Related FR
Doctor dashboard	Waiting reports are listed with the most important values and urgent wording first.	FR-58

11. Requirements Traceability Matrix

The matrix links each requirement to its use case and to the test that checks it. This gives a direct chain: SRS, use case, design, test case.

Test cases marked with a test name are automated pytest functions in the backend/tests folder (108 tests in total); they are run with the command pytest. A test case that says 'Manual test' is checked by hand.

Req. ID	Requirement	Use Case	Test Case
FR-01	The system shall allow registered users to log in with email and password.	UC-01	TC-01: test_login_me_and_invalid_login
FR-02	The system shall validate the entered credentials and show a general error message, which does not reveal which part was wrong, for invalid credentials.	UC-01	TC-02: test_login_me_and_invalid_login
FR-03	The system shall issue a signed token at login and shall refuse every private request that has no valid token.	UC-01	TC-03: test_private_endpoints_need_a_token
FR-04	The system shall allow users to log out, which ends the session in the browser.	UC-01	TC-04: Manual test: log out and press Back
FR-05	The system shall have no public sign-up. The first administrator can be created only once, on an empty system.	UC-03	TC-05: test_first_admin_only_once
FR-06	The system shall refuse login, and stop honouring old tokens, for deactivated users.	UC-01, UC-03	TC-06: test_deactivated_user_cannot_log_in
FR-07	The system shall store passwords only as bcrypt hashes and shall force a password change at first login when the password was generated by the system.	UC-02	TC-07a: test_passwords_are_hashed TC-07b: test_generated_password_is_returned_once_and_must_be_changed
FR-08	The system shall let a logged-in user change their own password after entering the current one.	UC-02	TC-08: test_generated_password_is_returned_once_and_must_be_changed

Req. ID	Requirement	Use Case	Test Case
FR-09	The administrator shall be able to create doctors, employees and pharmacy staff, and the system shall show the generated password once.	UC-03	TC-09: test_only_admin_manages_staff_and_audit
FR-10	The administrator shall be able to manage departments (create and list).	UC-03	TC-10: Fixture in conftest.py creates a department through the API
FR-11	The administrator shall be able to activate or deactivate a user, but not their own account.	UC-03	TC-11a: test_admin_cannot_deactivate_self TC-11b: test_deactivated_user_cannot_log_in
FR-12	Employees and the administrator shall be able to register a patient (the login account is created together with the patient) and edit only basic information.	UC-04	TC-12: test_employee_cannot_change_clinical_fields
FR-13	The system shall allow searching patients by name, patient code, phone number or email.	UC-04	TC-13: test_patient_search
FR-14	The system shall limit every function to its roles and shall answer others with the message 'You do not have permission to access this information.'	UC-01	TC-14a: test_permission_message TC-14b: test_role_matrix_on_core_endpoints
FR-15	The system shall keep patient data private: a patient sees only their own data, a doctor only authorised patients, and employees and the administrator never see clinical values.	UC-13, UC-14	TC-15a: test_patient_sees_only_own_records TC-15b: test_doctor_sees_only_authorized_patients TC-15c: test_employee_sees_basic_info_but_never_clinical_data TC-15d: test_admin_has_no_clinical_access
FR-16	The administrator shall be able to view a readable list of all people in the system.	UC-03	TC-16: test_people_list_and_readable_audit_labels
FR-17	The system shall accept PDF, JPG, JPEG, PNG and WEBP files (and optionally TXT and CSV) up to 10 MB, and shall check the real content type, not only the file name.	UC-06	TC-17: test_unsupported_and_fake_files_rejected
FR-18	The system shall read the text of text-based PDFs directly, without OCR.	UC-06	TC-18: test_text_pdf_is_read_without_ocr

Req. ID	Requirement	Use Case	Test Case
FR-19	The system shall use OCR for scanned PDFs and images, and shall show a clear message when OCR is not available.	UC-06	TC-19a: test_scanned_pdf_uses_ocr_pathway TC-19b: test_image_uses_ocr_pathway TC-19c: test_ocr_unavailable_error TC-19d: test_scanned_pdf_without_ocr_gives_clear_error
FR-20	The system shall extract test name, value, unit and reference range from the real file, whatever the layout, and shall keep the original test name.	UC-06	TC-20a: test_extracts_value_unit_reference TC-20b: test_real_world_layouts TC-20c: test_different_layout_and_missing_reference
FR-21	The system shall take reference ranges only from the report (forms a-b, below x, above x) and shall never invent a range.	UC-06	TC-21a: test_reference_parsing TC-21b: test_reference_none_when_absent
FR-22	The system shall decide Within, Above, Below or Unknown by fixed rules, and Unknown when no range is printed.	UC-06	TC-22a: test_range_classification TC-22b: test_less_than_classification TC-22c: test_greater_than_classification TC-22d: test_unknown_when_no_reference
FR-23	The system shall reject unreadable files and non-medical documents with a clear message, and shall save nothing in that case.	UC-06	TC-23a: test_invalid_document_rejected TC-23b: test_failed_or_invalid_upload_is_not_saved
FR-24	The system shall save an accepted report as 'Waiting for doctor' and show it on the assigned doctor's dashboard.	UC-06, UC-17	TC-24: test_complete_hospital_workflow
FR-25	The system shall warn when the name printed on the report does not match the selected patient.	UC-06	TC-25: test_name_mismatch_warning_and_report_values_come_from_each_file
FR-26	A doctor shall be able to upload a report for an authorised patient; when the patient has no assigned doctor the employee must choose one.	UC-06	TC-26a: test_doctor_can_upload_for_authorized_patient TC-26b: test_doctor_must_be_chosen_when_patient_has_none
FR-27	The system shall write explanations that do not diagnose or prescribe, pass them through a safety filter, and fall back to built-in text when no AI model is configured.	UC-06	TC-27a: test_disclaimer_and_safety_filter TC-27b: test_built_in_texts_never_diagnose_or_prescribe
FR-28	The system shall keep original files private and give them only to authorised users.	UC-08, UC-13	TC-28a: test_complete_hospital_workflow TC-28b: test_admin_has_no_clinical_access

Req. ID	Requirement	Use Case	Test Case
FR-29	The system shall accept other medical documents (imaging, ECG, pathology, discharge summary, prescription, clinical note), detect their type, and reject non-medical documents such as invoices.	UC-06	TC-29a: test_report_type_is_detected TC-29b: test_all_kinds_of_files_are_accepted_and_invoice_is_not TC-29c: test_non_medical_documents_are_still_rejected
FR-30	For every report the system shall produce key findings (normal or worth a look, with negation such as 'no evidence of' handled), red flags for urgent wording, suggestions for the doctor and general tips for the patient.	UC-07	TC-30a: test_statement_classification TC-30b: test_urgent_wording_is_flagged_for_the_doctor TC-30c: test_xray_gives_findings_and_suggestions_for_doctor_and_patient TC-30d: test_normal_ecg_has_no_attention_items
FR-31	For laboratory reports the insights shall be grouped by body system and use only printed ranges; a value without a printed range shall not be judged.	UC-07	TC-31a: test_lab_insights_group_by_body_system_and_use_only_printed_ranges TC-31b: test_value_without_printed_range_is_not_judged
FR-32	Doctor-only content (suggestions, red flags) shall be shown only to the doctor; patients shall see general tips only after the doctor's review; employees and administrators shall see none.	UC-07, UC-13	TC-32a: test_any_report_workflow_and_role_visibility TC-32b: test_lab_report_insights_visible_to_doctor_and_patient_tips_after_review
FR-33	A doctor shall be able to regenerate the insights of a report (recorded in the audit log); older reports without saved insights shall get them when opened.	UC-07	TC-33a: test_refresh_insights_is_doctor_only TC-33b: test_old_lab_reports_without_saved_insights_still_get_them
FR-34	An optional AI model may improve the wording only after personal details are removed; its answer shall pass the safety filter, otherwise the rule-based text is used.	UC-07	TC-34a: test_redaction_removes_personal_details TC-34b: test_ai_output_is_filtered_and_personal_data_is_not_sent TC-34c: test_bad_ai_reply_falls_back_to_rules TC-34d: test_doctor_filter_blocks_doses_and_patient_filter_blocks_diagnosis
FR-35	The doctor shall be able to review a report (approve, modify or ask for clarification); a comment is required for modify and clarify; the report is then marked Reviewed.	UC-08	TC-35a: test_complete_hospital_workflow TC-35b: test_doctor_content_validation
FR-36	The system shall show a report's analysis to the patient only after the doctor has reviewed it.	UC-08, UC-13	TC-36: test_unreviewed_analysis_hidden_from_patient_until_review

Req. ID	Requirement	Use Case	Test Case
FR-37	The doctor shall be able to add notes (Normal, Important, Follow-up required) and recommendations (with an optional follow-up date); patients shall not be able to edit them.	UC-09	TC-37a: test_patient_cannot_edit_doctor_content TC-37b: test_doctor_content_validation
FR-38	The doctor shall be able to add medical-history entries and shall see the patient's full history before reviewing.	UC-09	TC-38: test_doctor_sees_full_history_before_reviewing
FR-39	Only doctors shall create prescriptions (medicine, dosage, frequency, duration, quantity); a doctor shall see only their own prescriptions.	UC-10	TC-39: test_doctor_sees_only_their_own_prescriptions
FR-40	Pharmacy staff shall see prescriptions (cancelled ones are hidden) with only the data they need, and shall never create or edit a prescription.	UC-11	TC-40: test_pharmacy_sees_only_open_prescriptions_with_limited_data
FR-41	Pharmacy staff shall be able to dispense in part or in full; the system shall not allow more than the remaining quantity, shall block cancelled or expired prescriptions, and shall lower stock for tracked medicines (earliest expiry first, never expired stock) and refuse when the stock is not enough.	UC-11	TC-41a: test_expired_prescription_cannot_be_dispensed TC-41b: test_inventory_and_stock_deduction
FR-42	Pharmacy staff shall be able to mark a medicine as unavailable, and the system shall record it.	UC-11	TC-42: test_unavailable_medicine_is_recorded
FR-43	Pharmacy staff shall be able to manage medicine stock, and the dashboard shall show low-stock and soon-to-expire items.	UC-12	TC-43a: test_inventory_and_stock_deduction TC-43b: test_dashboards_have_real_numbers
FR-44	The patient shall be able to see the status of each prescription and how much has been dispensed.	UC-13	TC-44: test_complete_hospital_workflow

Req. ID	Requirement	Use Case	Test Case
FR-45	The system shall let the administrator, employees and patients book appointments, list them per role, and update their status (the doctor, employee and administrator may confirm, complete or cancel; a patient may only cancel); it shall refuse past times and clashes.	UC-05	TC-45: test_appointment_flow
FR-46	The system shall build a searchable patient timeline from the database (appointments, reports, analysis, reviews, notes, recommendations, prescriptions, dispensing, history) and show the source of each item.	UC-14	TC-46a: test_complete_hospital_workflow TC-46b: test_doctor_sees_full_history_before_reviewing
FR-47	The system shall compare two reports of one patient (previous, latest, change) with a trend that understands the direction of each parameter.	UC-15	TC-47a: test_comparison_of_two_reports TC-47b: test_comparison_trend_logic_units
FR-48	The system shall draw trend charts of each parameter over time.	UC-15	TC-48: test_trend_series
FR-49	A patient's comparison shall use only reviewed reports.	UC-15	TC-49: test_patient_comparison_only_uses_reviewed_reports
FR-50	Each role shall have a dashboard computed from the database, with honest empty states.	UC-17	TC-50: test_dashboards_have_real_numbers
FR-51	The system shall show in-app notifications on the dashboards: reports waiting for review, upcoming appointments, low stock and expiring medicine. (Email and SMS are not part of this version.)	UC-17	TC-51: test_dashboards_have_real_numbers
FR-52	The system shall write important actions (login, creation, upload, analysis, review, notes, prescriptions, opening a report file or a prescription, dispensing, user changes) to an audit log that only the administrator can read, in plain sentences.	UC-16	TC-52a: test_complete_hospital_workflow TC-52b: test_people_list_and_readable_audit_labels

Req. ID	Requirement	Use Case	Test Case
FR-53	The system shall display a medical disclaimer on every page and in the health response of the API.	UC-17	TC-53: test_disclaimer_and_safety_filter
FR-54	The system shall upgrade an existing database automatically at start-up when new columns are added, keeping the data.	UC-17	TC-54: test_new_columns_are_added_to_an_existing_database
FR-55	For a laboratory report the system shall show, at the top of the report, the number of tests read, in range, outside the range and without a range, and a card for each value outside the printed range with its difference from the limit, largest first.	UC-07	TC-55: Manual test: open a lab report as a doctor
FR-56	For other reports (Holter, ECG, scans and so on) the system shall read the label and number pairs of the text (counts, measurements, values with a printed range) and show them as tiles; it shall mark a tile only when the report prints a range for it or when it counts events above zero, shall leave out personal fields such as age, date or phone, and shall show each number once.	UC-07	TC-56a: test_table_rows_become_numbers_not_broken_sentences TC-56b: test_extract_metrics_ignores_ids_and_repeats
FR-57	The key findings of a report shall be real sentences only: an abbreviation such as 'No.' shall not end a sentence, table rows shall not appear as sentences, and a repeated sentence shall appear once.	UC-07	TC-57: test_table_rows_become_numbers_not_broken_sentences
FR-58	The doctor's dashboard shall show how many waiting reports have urgent wording and how many values are outside the range, and a 'look at these first' list of the waiting reports with the most important values pulled out, ordered by urgent wording and then by values outside the range.	UC-17	TC-58: Manual test: doctor dashboard with waiting reports

12. Appendices

Appendix A: Glossary

The terms used in this document are defined in section 1.6. Further terms:

Term	Meaning
Within / Above / Below / Unknown	The four statuses of a result: inside the printed range, higher than it, lower than it, or no range printed.
Waiting for doctor	A saved report that the doctor has not reviewed yet; the patient cannot see its analysis.
Reviewed	A report the doctor has approved, modified or clarified; it becomes visible to the patient.
Safety filter	A check that blocks diagnosis, doses and other unsafe wording in automatic text.
Audit log	A permanent record of important actions, readable by the administrator.
Patient code	A readable patient number printed on registration, for example for searching.

Appendix B: Sample Forms and Screens

The screens below were captured from the running application with demonstration data (the fictional patient Rahul Patil and demo staff accounts). The list of screens is in section 5.1.

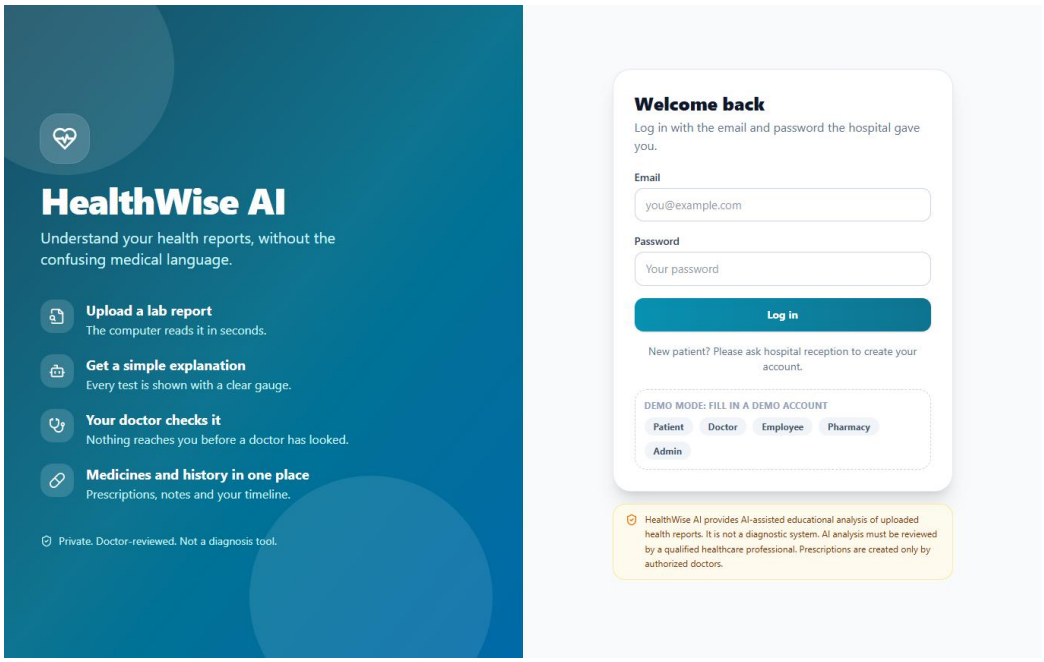


Figure B.1 Login page

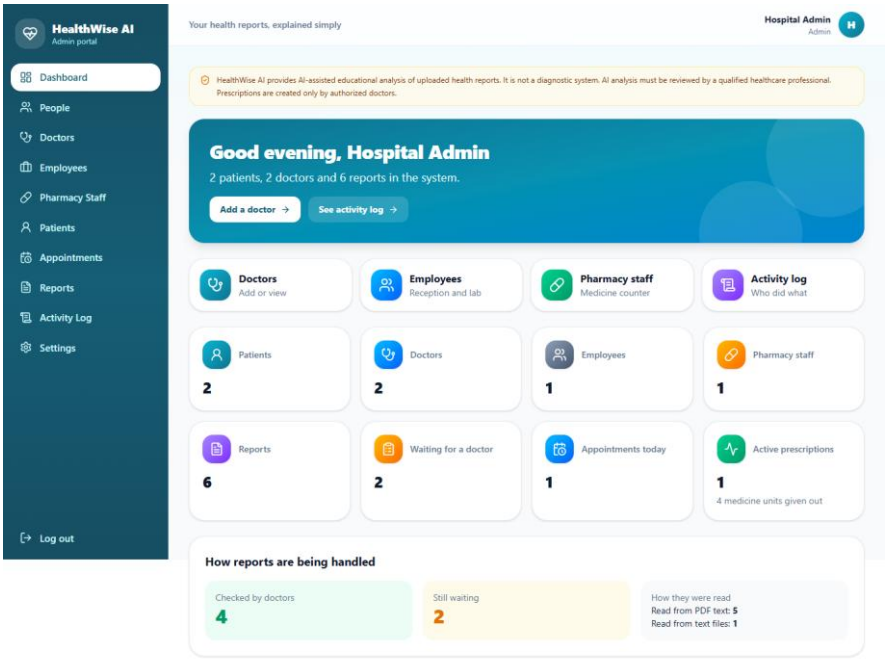


Figure B.2 Admin dashboard

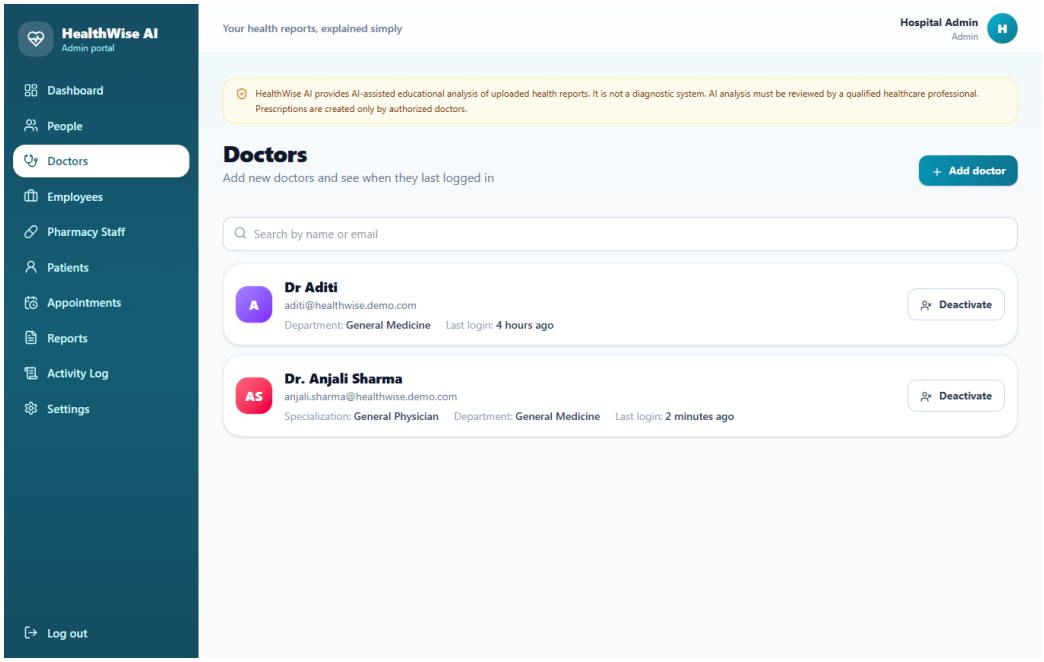


Figure B.3 Admin: doctors list (staff management)

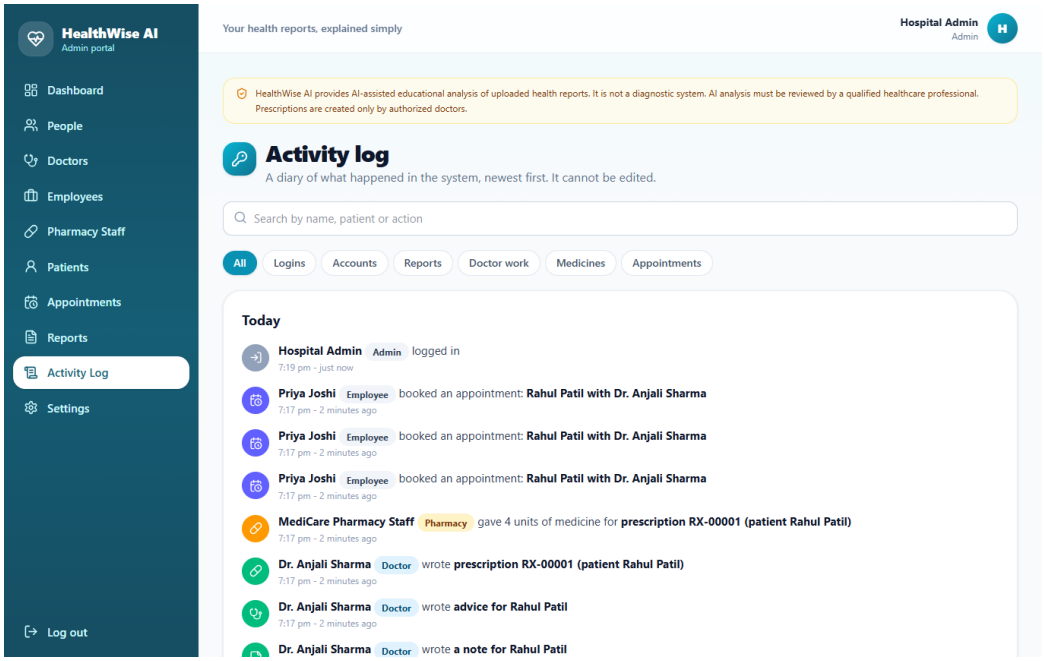


Figure B.4 Admin: audit log

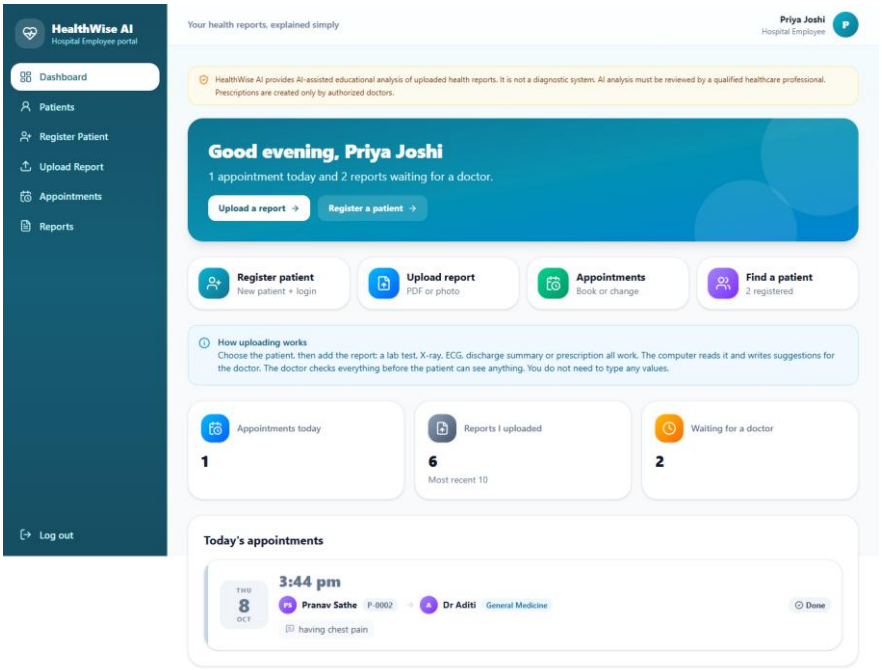


Figure B.5 Employee dashboard

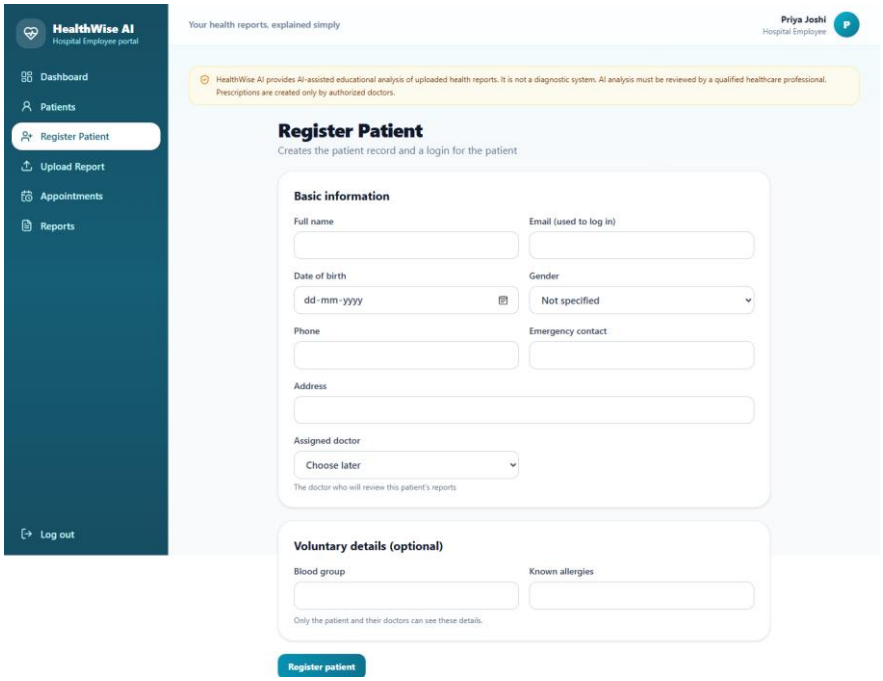


Figure B.6 Employee: register patient

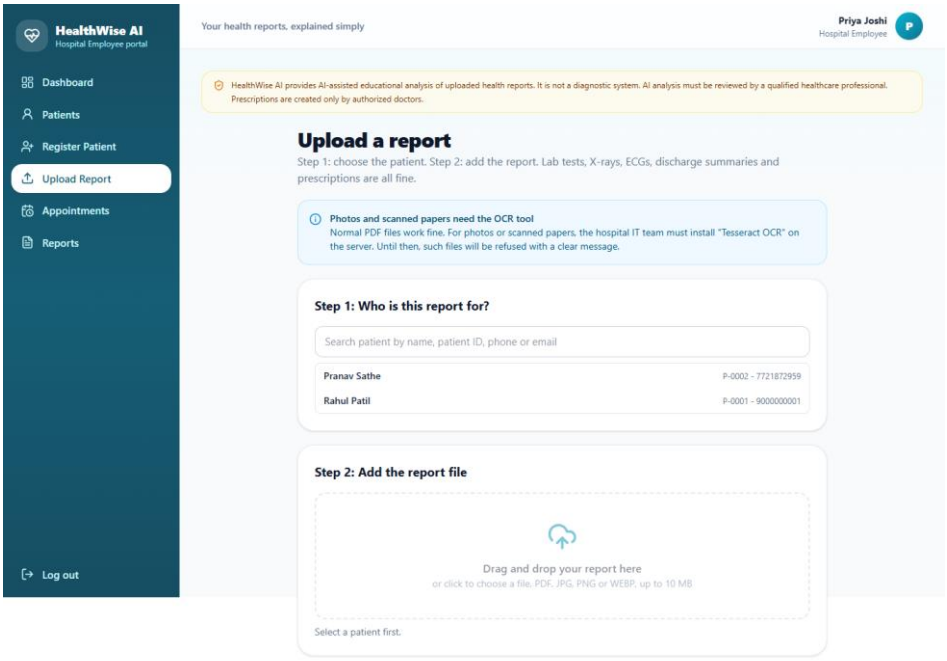


Figure B.7 Employee: upload report

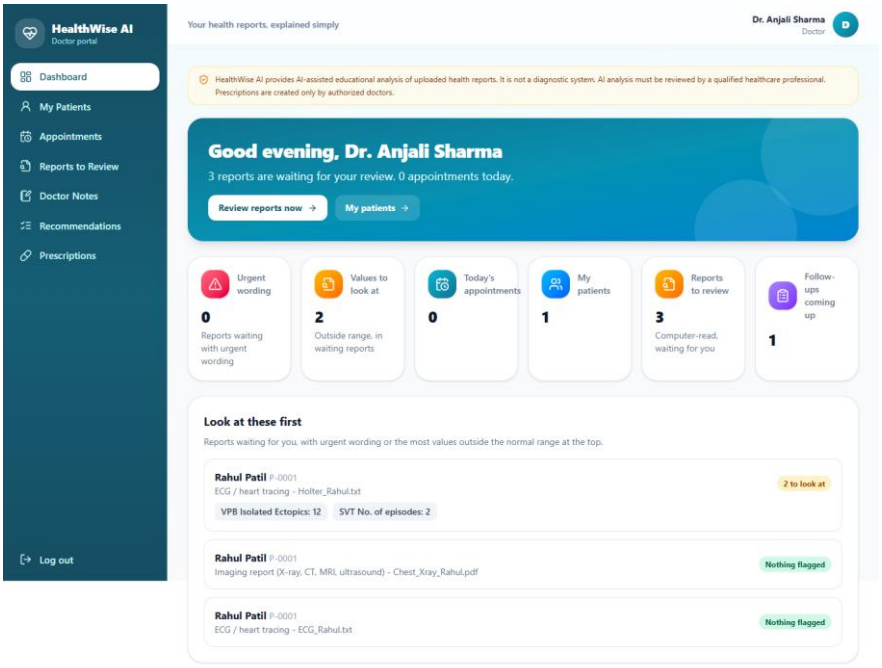


Figure B.8 Doctor dashboard

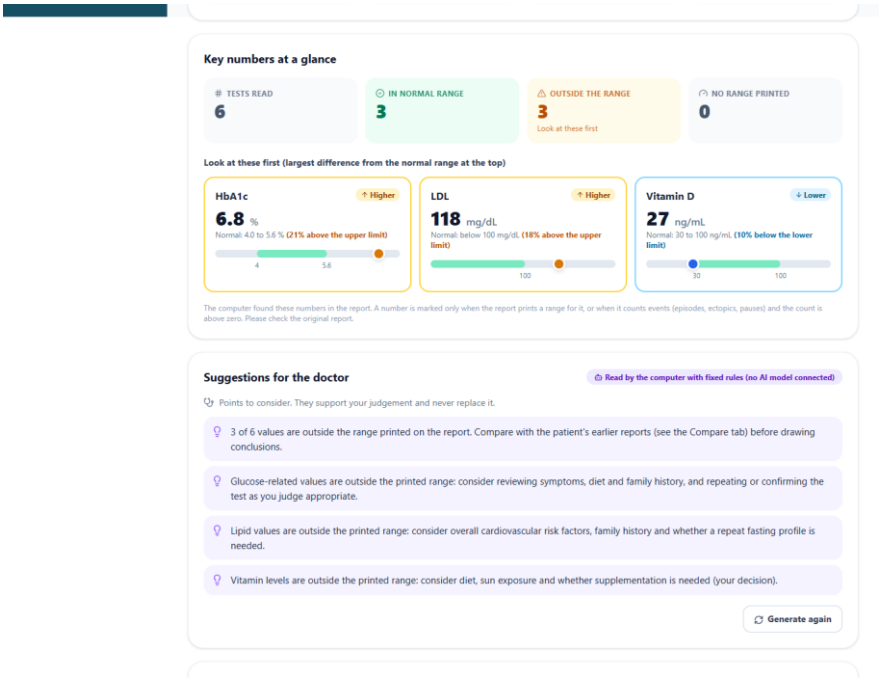


Figure B.9 Doctor: report review, key numbers of a laboratory report

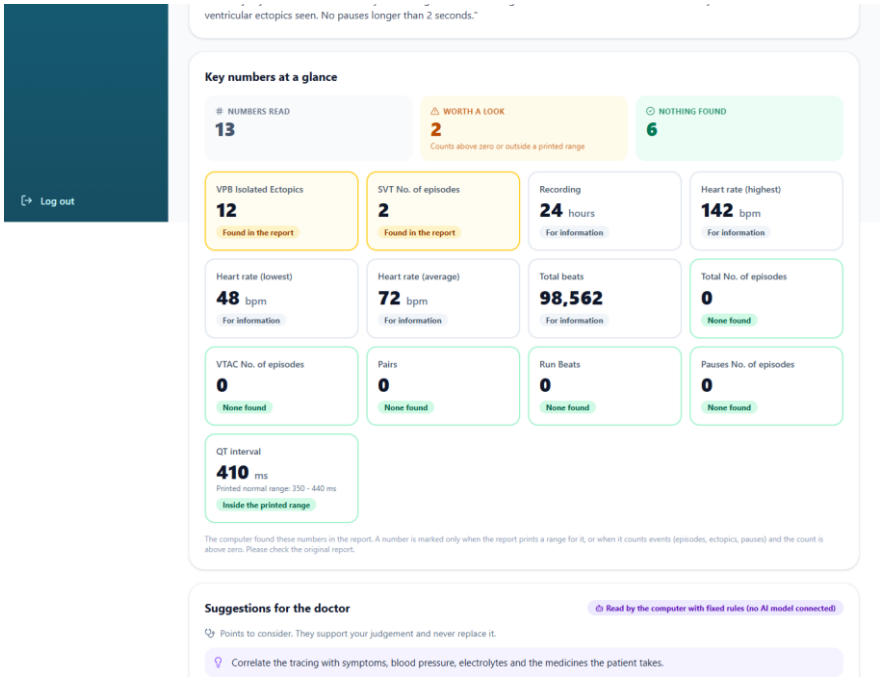


Figure B.10 Doctor: report review, key numbers of a Holter report

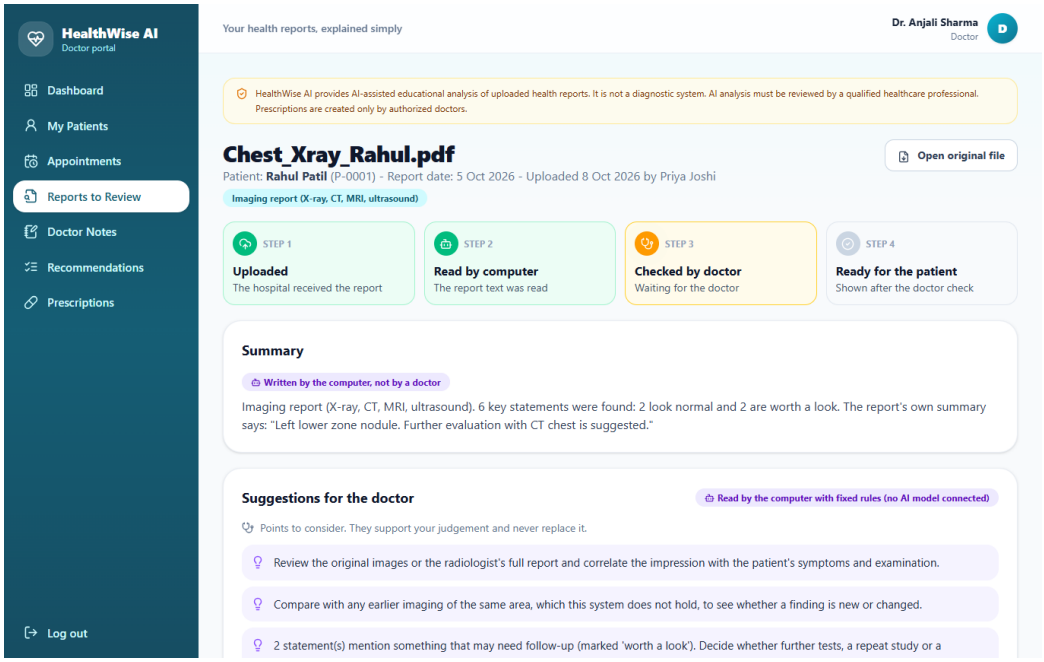


Figure B.11 Doctor: report review (chest X-ray, insights for the doctor)

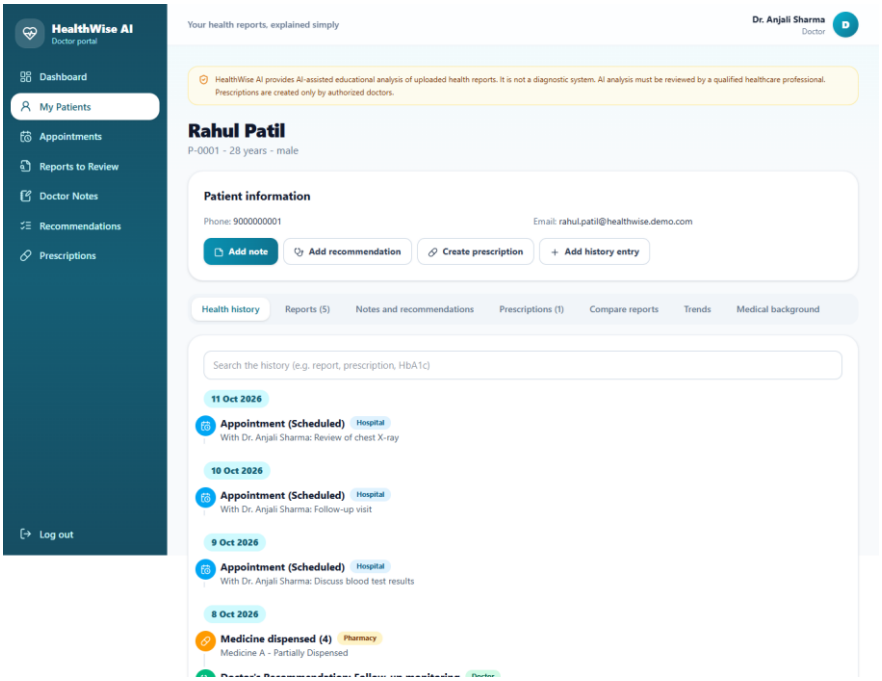


Figure B.12 Doctor: patient details

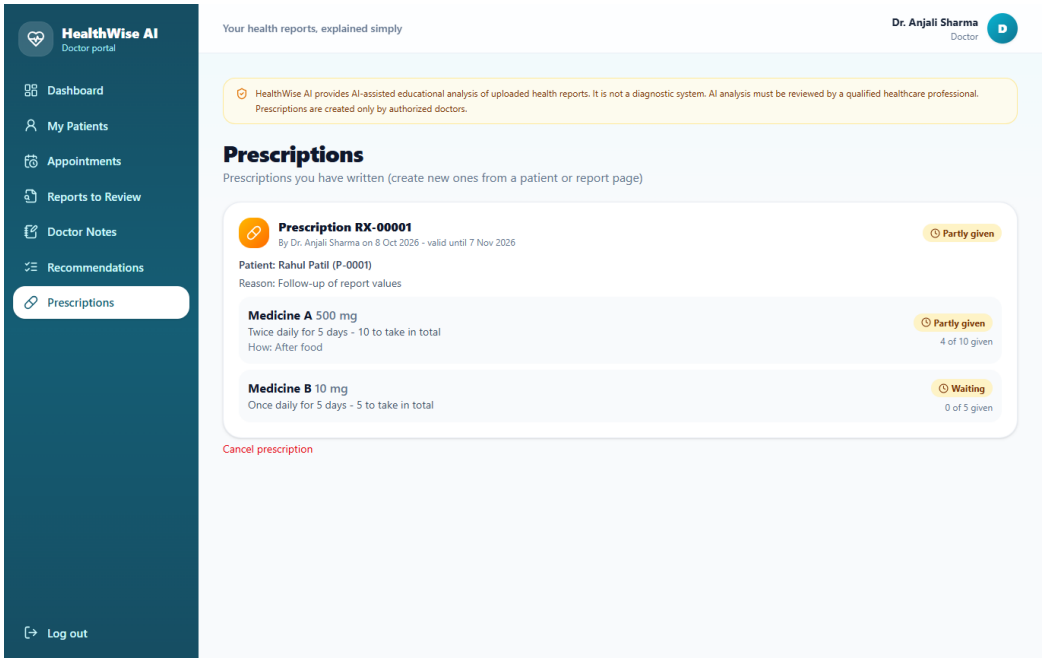


Figure B.13 Doctor: prescriptions

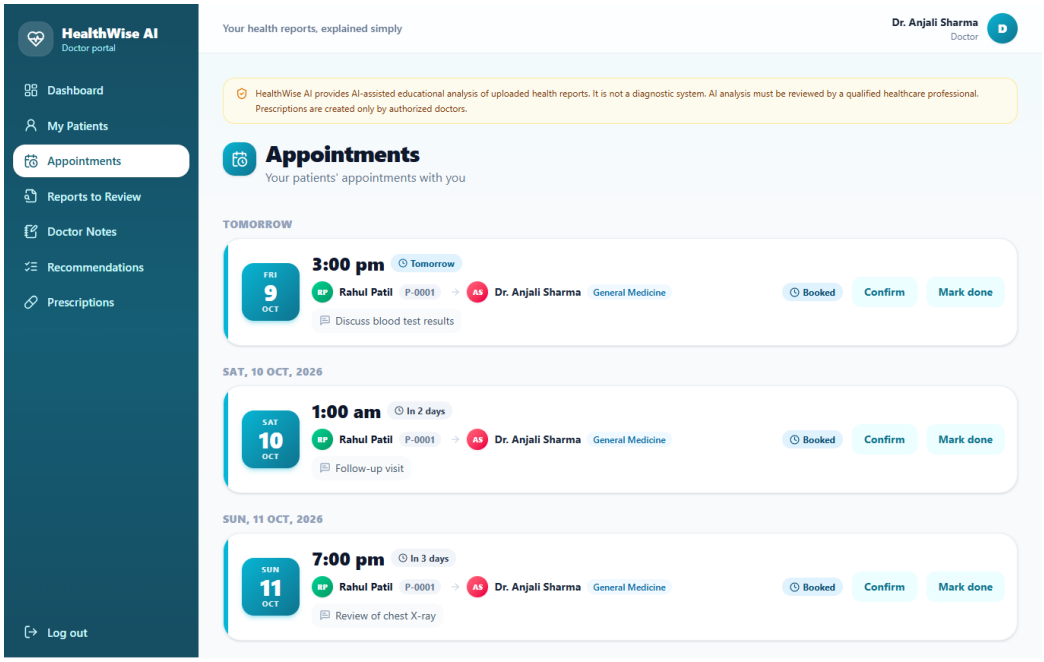


Figure B.14 Appointments

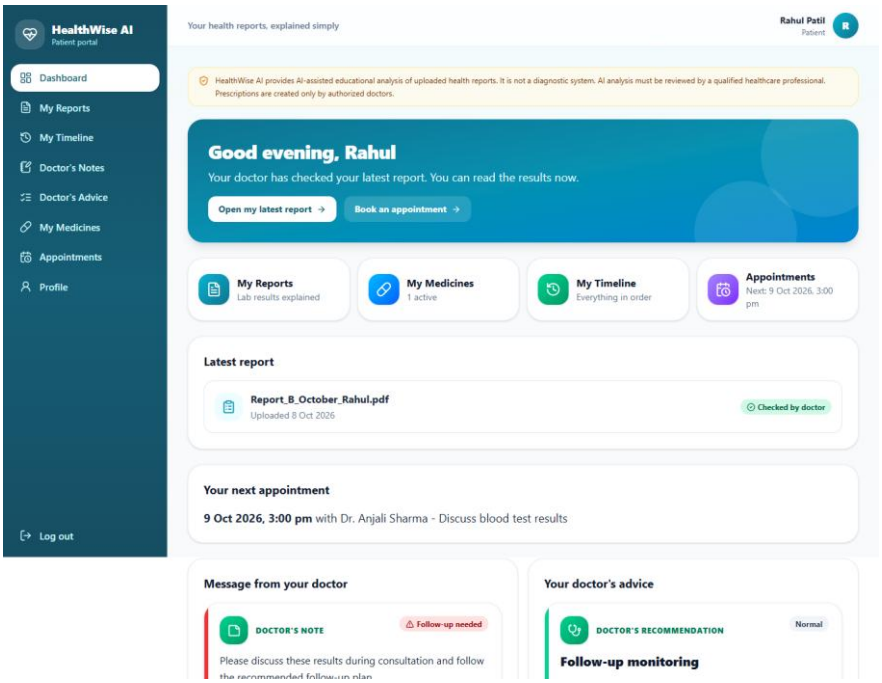


Figure B.15 Patient dashboard

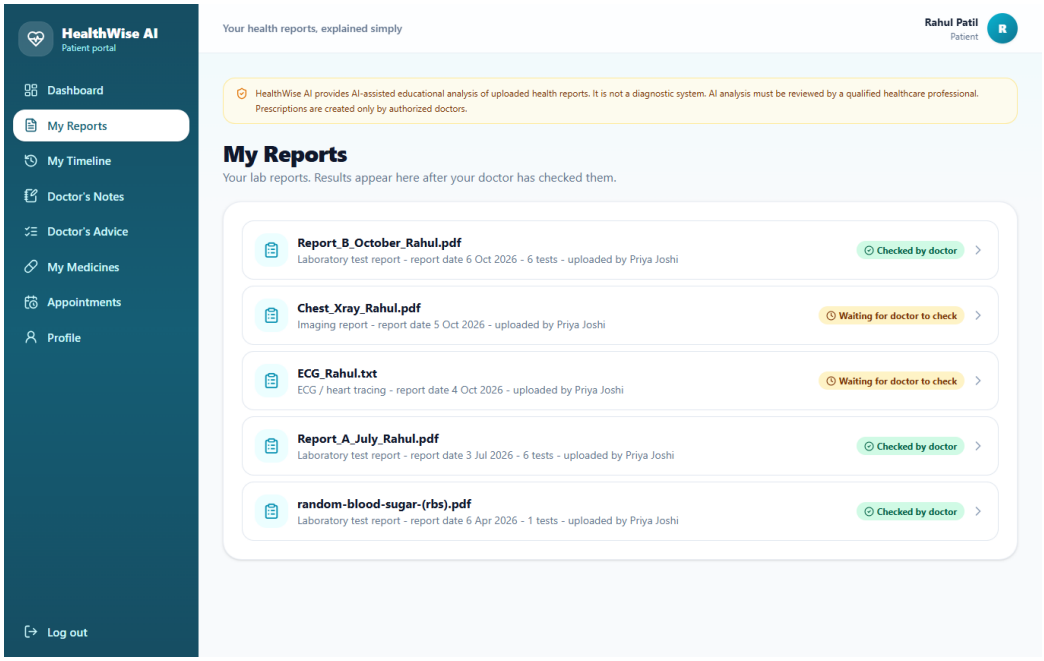


Figure B.16 Patient: my reports

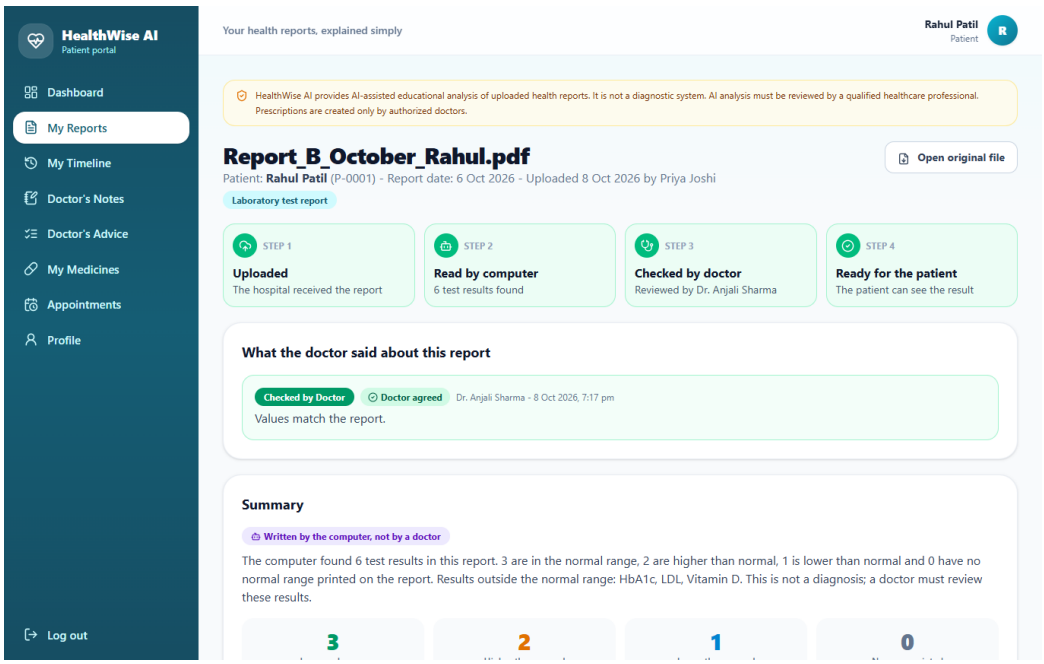


Figure B.17 Patient: reviewed report with tips

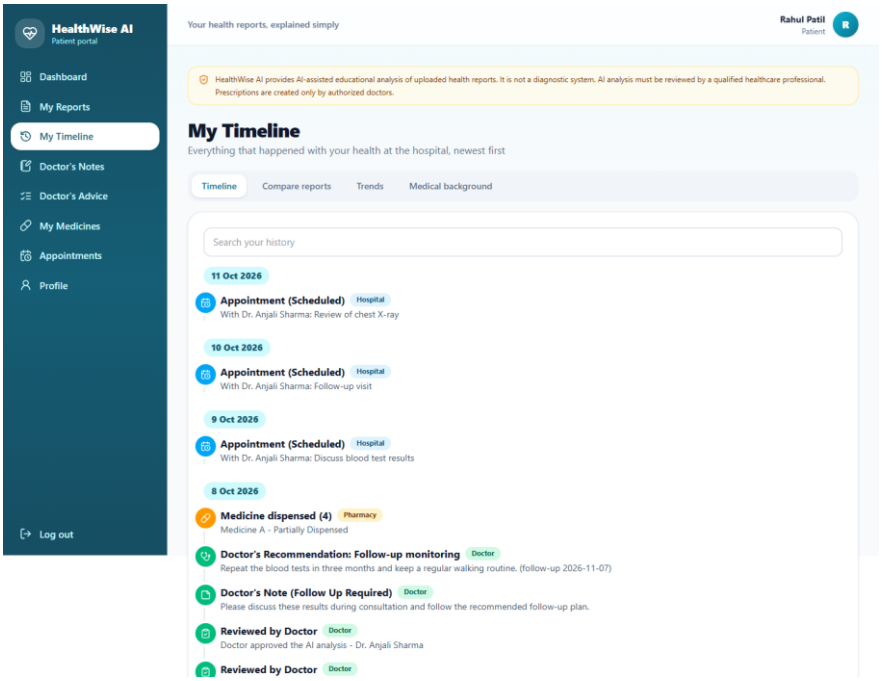


Figure B.18 Patient: health timeline

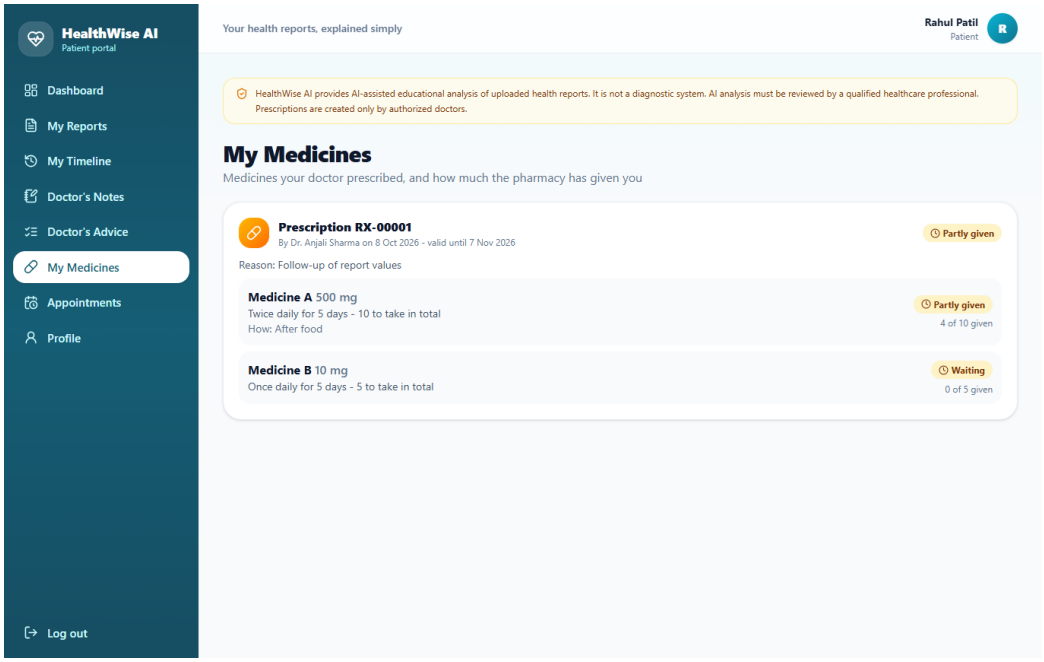


Figure B.19 Patient: my medicines

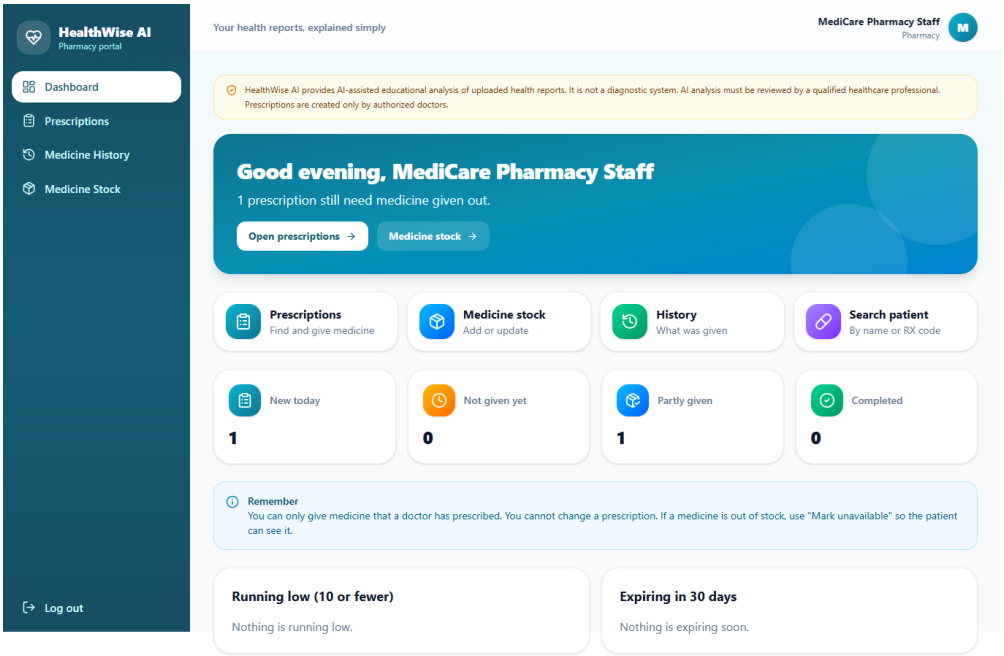


Figure B.20 Pharmacy dashboard

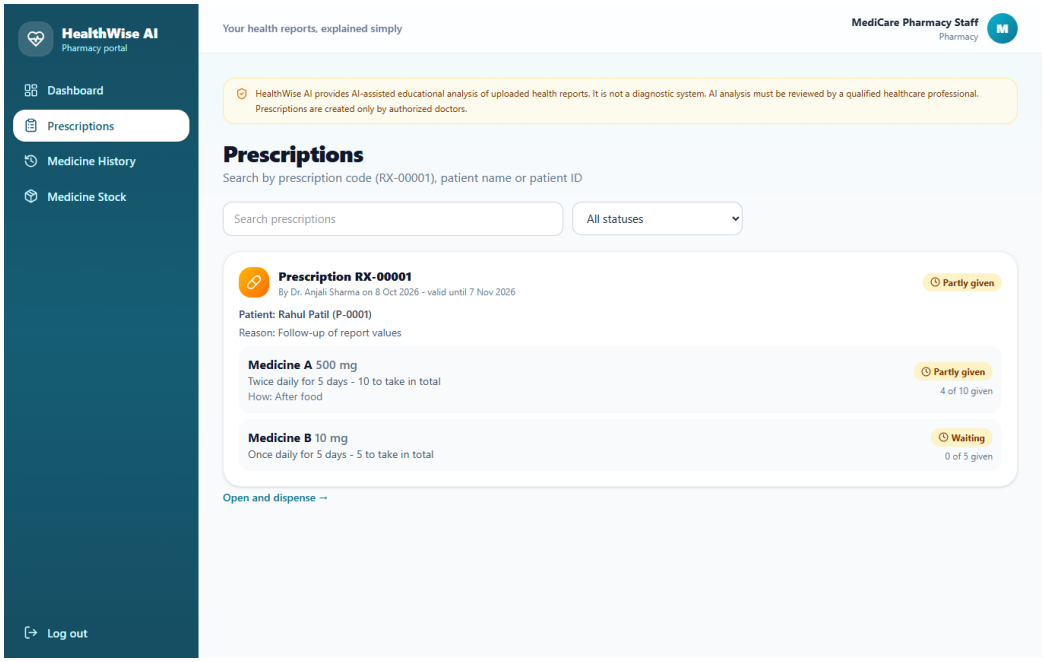


Figure B.21 Pharmacy: prescriptions to dispense

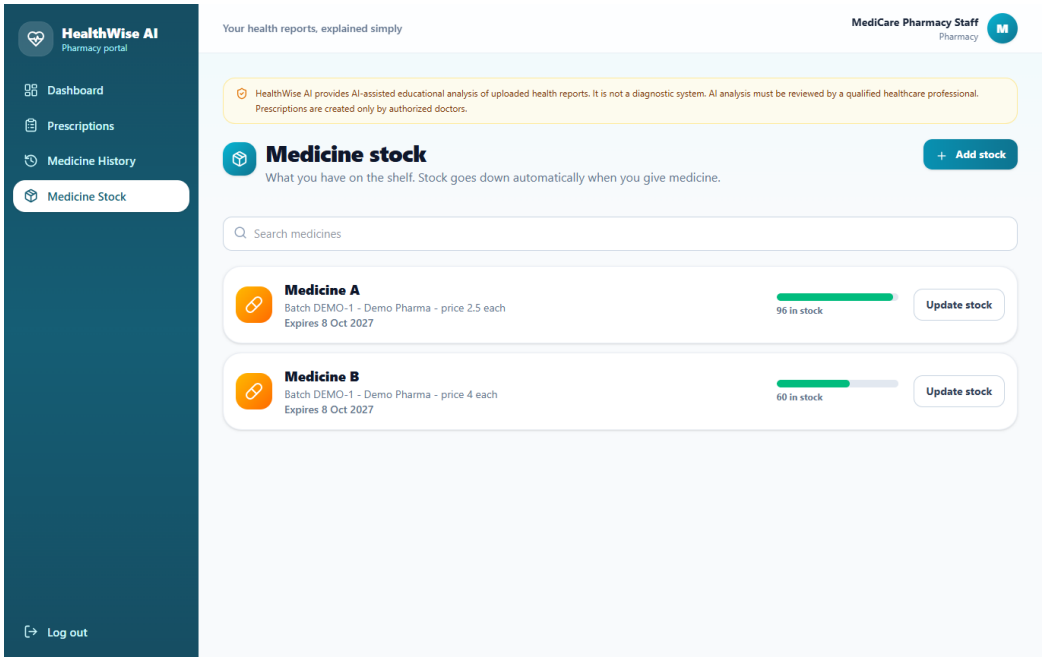


Figure B.22 Pharmacy: medicine stock

Appendix C: Sample Reports

The folder sample_reports in the project holds files for testing and for the demonstration.

File	What it shows
Report_A_July_Rahul.pdf	Text PDF, blood and metabolic panel (earlier report).
Report_B_October_Rahul.pdf	Text PDF, same patient three months later (for comparison and trends).
Report_A_scan.png	Scanned image; needs OCR.
Report_C_other_patient.txt	Different patient, different layout and values (proves values come from each file).
Chest_Xray_Rahul.pdf	Imaging report with findings and impression.
ECG_Rahul.txt	ECG report.
Discharge_Summary_Rahul.pdf	Discharge summary.
Prescription_Rahul.txt	Prescription document.
Not_a_health_report.txt	Non-medical file; must be rejected.

Appendix D: Role-Permission Matrix

Function	Admin	Doctor	Employee	Patient	Pharmacy
Manage staff and departments	Yes	-	-	-	-
Read audit log	Yes	-	-	-	-
Register patient	Yes	-	Yes	-	-

Function	Admin	Doctor	Employee	Patient	Pharmacy
Upload report	-	Authorised patients	Yes	-	-
See clinical values and insights for doctors	-	Authorised patients	-	-	-
Review a report, write notes and advice	-	Yes	-	-	-
See own reviewed reports and tips	-	-	-	Yes	-
Create a prescription	-	Yes	-	-	-
Dispense medicine, manage stock	-	-	-	-	Yes
Book appointments	Yes	-	Yes	Yes	-
Change appointment status	Yes	Yes	Yes	Cancel only	-

Appendix E: Database Schema

Chapter 8 lists every table and column; Figures 7.8 and 7.9 show the main relationships. The foreign keys of the database are:

Table	Column	Refers to
doctors	user_id	users . id
doctors	department_id	departments . id
employees	user_id	users . id
pharmacy_staff	user_id	users . id
patients	user_id	users . id
patients	assigned_doctor_id	doctors . id
appointments	patient_id	patients . id
appointments	doctor_id	doctors . id
appointments	department_id	departments . id
appointments	created_by	users . id
health_reports	patient_id	patients . id
health_reports	uploaded_by	users . id
health_reports	assigned_doctor_id	doctors . id
health_reports	reviewed_by	doctors . id
health_parameters	report_id	health_reports . id
report_comparisons	patient_id	patients . id
report_comparisons	previous_report_id	health_reports . id
report_comparisons	latest_report_id	health_reports . id
doctor_notes	patient_id	patients . id
doctor_notes	doctor_id	doctors . id

Table	Column	Refers to
doctor_notes	report_id	health_reports . id
doctor_recommendations	patient_id	patients . id
doctor_recommendations	doctor_id	doctors . id
doctor_recommendations	report_id	health_reports . id
medical_history	patient_id	patients . id
medical_history	doctor_id	doctors . id
prescriptions	patient_id	patients . id
prescriptions	doctor_id	doctors . id
prescriptions	report_id	health_reports . id
prescription_items	prescription_id	prescriptions . id
medicine_dispensing	prescription_id	prescriptions . id
medicine_dispensing	prescription_item_id	prescription_items . id
medicine_dispensing	pharmacy_staff_id	pharmacy_staff . id
audit_logs	user_id	users . id

The complete SQL script is in backend/schema.sql of the project.

Appendix F: Sample Questionnaire

This is a sample form for the user-acceptance survey. It has not been used yet, and no answers are reported in this document.

No.	Statement	1 (disagree) ... 5 (agree)
1	The screens were easy to understand without any training.	1 2 3 4 5
2	I could find the task I wanted from the dashboard in a few clicks.	1 2 3 4 5
3	The result cards and colours made the report easy to understand.	1 2 3 4 5
4	The suggestions for the doctor saved time when reviewing a report.	1 2 3 4 5
5	The tips for the patient were clear and did not frighten me.	1 2 3 4 5
6	Error messages told me what to do next.	1 2 3 4 5
7	I felt my information was kept private.	1 2 3 4 5
8	Overall, I would like to use this system in a hospital.	1 2 3 4 5

Appendix G: References

See section 1.7.